

GENERALIST REPRESENTATION, SPECIALIST DETECTION: TS-ROUTER FOR TIME-SERIES ANOMALY DETECTION

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ABSTRACT

Time-series anomaly detection (TSAD) is difficult to generalize across datasets because heterogeneous temporal dynamics imply different notions of normality and favor different detection criteria. While time-series foundation models provide transferable representations, coupling them with a fixed anomaly-scoring mechanism can overlook this variation. This motivates a different perspective on foundation-model-based TSAD: using foundation models to coordinate specialized anomaly criteria rather than directly imposing a universal one. Based on this view, we propose **TS-Router**, a generalist-representation, specialist-detection framework that estimates the relative competence of heterogeneous anomaly detectors from pretrained temporal representations and selects suitable specialists for each target series. To avoid relying on specialist-performance labels from real tasks, we derive soft competence supervision from specialists' relative performance on labeled simulated tasks. At deployment, routing requires no target anomaly labels, and only the selected specialists are fitted unsupervisedly on the target series. We bound Top- k set-competence regret under representation coverage and conditional competence stability. Across 16 real-world benchmarks and four complementary evaluation metrics, TS-Router achieves the best overall average rank. Controlled ablations with multiple frozen TSFM encoders further support the use of pretrained representations for competence estimation and adaptive specialist selection. The code is available at <https://anonymous.4open.science/r/TS-Router-D8FF>.

1 INTRODUCTION

Time-series anomaly detection (TSAD) is a fundamental task in sequential data analysis, with broad applications in industrial monitoring (Nizam et al., 2022; Chen et al., 2022), system operations (Audibert et al., 2020; Guo et al., 2024), health care (Su et al., 2019; Xu et al., 2021), and transportation and energy systems (Deng & Hooi, 2021). TSAD aims to identify observations or subsequences that are inconsistent with the normal behavior of the underlying temporal process, typically in settings

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where anomalies are rare and reliable anomaly annotations are unavailable. Accordingly, reliable TSAD requires two coupled capabilities: modeling the contextual regularities that characterize normal temporal behavior, and assessing whether deviations from these regularities indicate anomalies or merely reflect normal variability (Mueller, 2025; Wang et al., 2025).

Conventional TSAD methods learn normal behavior separately for each target series and score deviations from it. For example, PCA-based detectors fit a low-dimensional subspace and flag observations with large reconstruction errors (Yairi et al., 2001). This target-specific design leads to a one-dataset-one-model paradigm, requiring a detector to be selected, tuned, and fitted for every target. Yet detector suitability varies across series, since different temporal dynamics favor different anomaly criteria. Detector selection therefore depends on domain expertise and repeated trial and error. Automated selection methods alleviate this burden by choosing among candidate detectors based on target characteristics (Zhao et al., 2021; Goswami et al., 2022; Sylligardos et al., 2023; Schmidl et al., 2024). However, they typically rely on handcrafted descriptors, limited historical performance records, or deployment-time candidate evaluation. This exposes a complementary challenge: while detector suitability is target-dependent, inferring such suitability on an unseen series requires a representation of temporal structure that transfers across datasets.

Time-series foundation models (TSFMs) offer a natural way to address this representational challenge by learning transferable temporal representations from large-scale corpora that can be readily applied to new target series. This capability has motivated growing interest in applying models such as MOMENT and TimesFM to TSAD (Goswami et al., 2024; Das et al., 2024). Most existing TSFM-based approaches, however, couple these transferable representations with forecasting or reconstruction objectives and use the resulting error as a fixed anomaly criterion. A recent systematic evaluation shows that such direct adaptations may perform no better than simple local-statistics baselines across diverse time series (Zhu et al., 2026). Rather than negating the value of pretrained temporal representations, these findings suggest that solving the representation problem does not necessarily solve the anomaly-decision problem, raising a more fundamental question: **beyond directly defining an anomaly criterion, what role should foundation models play in TSAD?**

This question reflects a fundamental difference between forecasting and anomaly detection. Autoregressive forecasting admits a shared objective, such as $\mathbf{x}_{1:t} \mapsto \mathbf{x}_{t+1:t+H}$, where each observed continuation provides supervision, enabling pretraining on abundant unlabeled series. TSAD, in contrast, must determine which departures from normal behavior constitute anomalies, despite scarce annotations and anomaly criteria whose suitability varies across targets. The same pattern may be normal for one series but anomalous for another, and different detectors may capture different forms of deviation. As shown in Figure 1, the relative strengths of representative specialists and the anomaly-specific foundation model TimeRCD (Lan et al., 2025b) vary substantially across domains. Together, these observations motivate conditional specialization and suggest a separation between what can be transferred and what should remain adaptive: temporal representations may generalize across series, whereas the choice of anomaly criterion can benefit from target-dependent adaptation.

Motivated by this separation, we develop **TS-Router**. Given an unlabeled target series, a frozen foundation encoder represents its temporal context, and a learned router predicts the relative competence of candidate specialists. Only the selected detectors are then fitted unsupervisedly on the target series, and their normalized anomaly scores are combined for detection. This route-before-fit procedure avoids fitting every candidate at deployment and requires no target anomaly labels. Our main contributions are:

- We introduce a generalist-representation, specialist-detection perspective for TSAD, separating transferable temporal representation from target-dependent anomaly decision and positioning foundation models as coordinators of specialized detection criteria through competence routing.

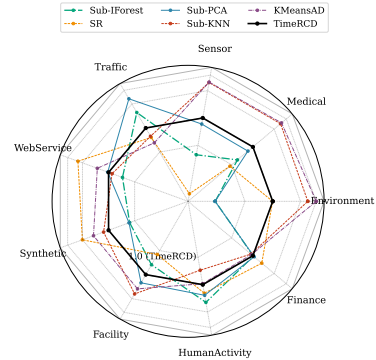


Figure 1: Domain-wise VUS-PR of representative specialists relative to TimeRCD, normalized by TimeRCD performance.

- We propose **TS-Router**, which uses transferable temporal representations to estimate heterogeneous detector competence and adaptively select and combine suitable specialists. Without large-scale real anomaly annotations, we derive scalable soft routing supervision from specialists’ relative performance on labeled simulated series, and establish a Top- k set-competence regret bound under explicit transfer assumptions.
- Extensive experiments on diverse TSAD benchmarks show that TS-Router achieves the best overall average rank among the evaluated baselines. Controlled routing and representation ablations demonstrate the benefits of adaptive specialist selection and show that the framework is effective with multiple frozen TSFM encoders.

2 RELATED WORK

Target-Specific Time-Series Anomaly Detection. Most unsupervised TSAD methods are fitted separately to each target series and identify anomalies as deviations from target-specific normality. Deep approaches commonly use reconstruction objectives, as in USAD, OmniAnomaly, TranAD, and TFMAE (Audibert et al., 2020; Su et al., 2019; Tuli et al., 2022; Fang et al., 2024), or learn discriminative representations through self-supervised and contrastive objectives (Yang et al., 2023; Xu et al., 2021; Shen et al., 2020; Darban et al., 2025b; Lan et al., 2025a). Classical detectors based on statistical, distance, density, subspace, isolation, and one-class criteria also remain competitive (Goldstein & Dengel, 2012; Li et al., 2007; Ramaswamy et al., 2000; Breunig et al., 2000; Yairi et al., 2001; Schölkopf et al., 1999; Liu et al., 2008; Ren et al., 2019). These methods embody different inductive biases for modeling normality and scoring deviations, whose suitability can vary substantially across target series. Consequently, deployment still requires target-specific detector selection and adaptation, limiting scalable generalization across heterogeneous series.

Time-Series Foundation Models for Anomaly Detection. Time-series foundation models learn transferable temporal representations through large-scale pretraining. General-purpose models such as MOMENT and UniTS support multiple downstream tasks, while TimesFM, Chronos, and Time-MoE primarily transfer forecasting capabilities across datasets (Goswami et al., 2024; Gao et al., 2024; Das et al., 2024; Ansari et al., 2024; Shi et al., 2024). Pretrained models have also been adapted to anomaly detection, including DADA and TSPulse (Shentu et al., 2024; Ekambaram et al., 2025). More recently, TimeRCD develops an anomaly-specific foundation model pretrained with labeled synthetic data and achieves strong zero-shot performance (Lan et al., 2025b). Despite their transferable representations, these approaches generally couple them with a unified detector or fixed anomaly-scoring mechanism across heterogeneous targets. Direct use of forecasting or reconstruction errors may also perform comparably to simple local-statistics baselines, since anomalies are not consistently harder to forecast or reconstruct (Zhu et al., 2026). These observations leave open whether pretrained temporal representations can instead support target-dependent adaptation of anomaly criteria across heterogeneous series.

Automated Detector Selection. Automated detector-selection methods address detector heterogeneity by identifying suitable candidates for unlabeled targets. MetaOD transfers historical detector performance through meta-learning and handcrafted descriptors (Zhao et al., 2021), while Unsupervised Model Selection and Choose Wisely assess detector suitability using surrogate criteria, candidate outputs, or series characteristics (Sylligardos et al., 2023; Goswami et al., 2022). AutoTSAD further automates detector configuration, selection, and ensembling for individual targets (Schmidl et al., 2024). Meanwhile, pseudo-anomaly and anomaly-injection methods use synthetic samples primarily to train anomaly detectors directly (Obata et al., 2025; Jeong et al., 2023; Darban et al., 2025a; Shentu et al., 2024). Existing selection methods, however, remain tied to handcrafted descriptors, historical task collections, surrogate evaluations, or deployment-time candidate execution, limiting transferable and scalable competence estimation on unseen series.

3 PROBLEM FORMULATION

TSAD with Heterogeneous Specialists. Let $\mathcal{T} = (\mathbf{X}, \mathbf{y})$ denote a TSAD task, where $\mathbf{X} = (x_1, \dots, x_L) \in \mathbb{R}^L$ is the observed series and $\mathbf{y} \in \{0, 1\}^L$ contains its anomaly labels. We formulate the selection problem for univariate series; multivariate targets use variable-wise routing followed by score aggregation. Target labels are unavailable at deployment and are used only to de-

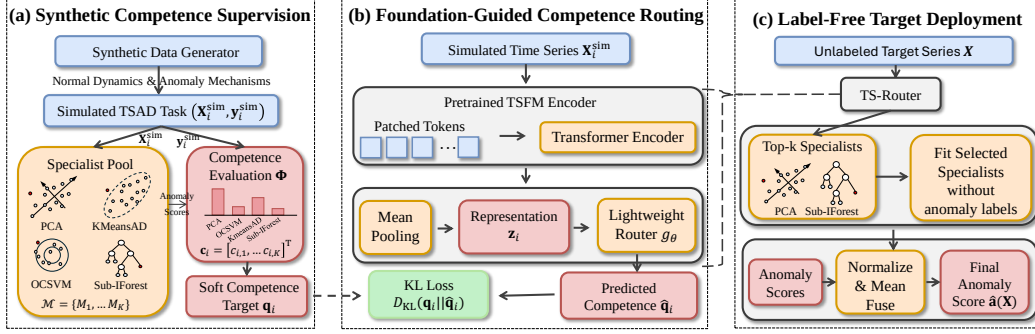


Figure 2: Overview of the TS-Router framework.

fine or evaluate competence. Consider a pool $\mathcal{M} = \{M_1, \dots, M_K\}$ of heterogeneous unsupervised detectors. Each M_j is a *specialist* with a particular inductive bias for modeling normality and scoring deviations, rather than a detector assigned to a predefined anomaly type. Under a fixed label-free fitting and scoring protocol, it produces scores $\mathbf{a}_j(\mathbf{X})$ with competence $c_j(\mathcal{T}) = \Phi(\mathbf{a}_j(\mathbf{X}), \mathbf{y})$. The utility Φ is evaluated on the protocol’s evaluation interval and expressed on the $[0, 1]$ scale. The vector $\mathbf{c}(\mathcal{T}) = [c_1(\mathcal{T}), \dots, c_K(\mathcal{T})]^\top$ records specialist suitability.

Specialist Selection with a Fixed Budget. For a budget $1 \leq k \leq K$, let $\mathcal{I}_k = \{S \subseteq [K] : |S| = k\}$, where $[K] = \{1, \dots, K\}$. We assess a selected set through its mean individual competence. For $\mathbf{v} \in \mathbb{R}^K$, write

$$V_k(S, \mathbf{v}) = \frac{1}{k} \sum_{j \in S} v_j, \quad \mathcal{T}_k(\mathbf{v}) = \max_{S \in \mathcal{I}_k} V_k(S, \mathbf{v}). \quad (1)$$

The maximizing set is $\text{TopK}(\mathbf{v}, k)$, with a fixed tie-breaking rule. The best globally fixed set and hindsight task-wise selection attain

$$U_{\text{fixed}}^{(k)} = \max_{S \in \mathcal{I}_k} \mathbb{E}[V_k(S, \mathbf{c}(\mathcal{T}))], \quad U_{\text{oracle}}^{(k)} = \mathbb{E}[\mathcal{T}_k(\mathbf{c}(\mathcal{T}))]. \quad (2)$$

These utilities compare sets of the same size before score fusion; they are not the detection utility of an averaged score sequence. When $k = 1$, they reduce to individual-specialist selection.

Representation-Conditioned Competence. Although $\mathbf{c}(\mathcal{T})$ is unobserved at deployment, temporal information can guide selection. Let $Z = E(\mathbf{X})$ and $\boldsymbol{\mu}(z) = \mathbb{E}[\mathbf{c}(\mathcal{T}) \mid Z = z]$. The best budget- k selection based on Z attains $U_{\text{cond}}^{(k)}(Z) = \mathbb{E}_Z[\mathcal{T}_k(\boldsymbol{\mu}(Z))]$, and

$$U_{\text{fixed}}^{(k)} \leq U_{\text{cond}}^{(k)}(Z) \leq U_{\text{oracle}}^{(k)}. \quad (3)$$

Thus, observable context can improve selection over a fixed set, while $U_{\text{oracle}}^{(k)} - U_{\text{cond}}^{(k)}(Z)$ measures the information gap to hindsight selection. If $Z_0 = h(Z_1)$, then $U_{\text{cond}}^{(k)}(Z_1) \geq U_{\text{cond}}^{(k)}(Z_0)$. Transferable representations therefore need not directly define a universal anomaly criterion: they can instead inform which specialist criteria to deploy.

Competence Routing. A router g_θ maps $z = E(\mathbf{X})$ to a predicted competence distribution $\hat{\mathbf{q}}_\theta(z) \in \Delta^{K-1}$. Its Top- k index set is $\hat{S}_\theta^{(k)}(z) = \text{TopK}(\hat{\mathbf{q}}_\theta(z), k)$, corresponding to the specialist set $\mathcal{S}_\theta(\mathbf{X}) = \{M_j : j \in \hat{S}_\theta^{(k)}(E(\mathbf{X}))\}$. Routing precedes detector fitting. Only these specialists are fitted without target labels, and their normalized scores are combined for detection. This separates representation-conditioned specialist selection from the subsequent use of their detection outputs.

4 METHOD

As illustrated in Figure 2, TS-Router learns specialist selection from temporal representations and synthetic soft competence supervision. Training follows two stages: a time-series foundation model first learns a temporal representation under its native objective; the encoder is then frozen, and a lightweight router is trained to predict specialist competence from the fixed representation. At deployment, routing precedes detector fitting, after which only the selected specialists are fitted

without anomaly labels and combined for detection. This separates general temporal representation learning from target-adaptive anomaly decision.

4.1 SYNTHETIC COMPETENCE SUPERVISION

Real-world TSAD data provide insufficient supervision for learning how specialist competence varies across temporal contexts, as anomaly labels are scarce and anomalies are rare. Following the labeled simulation setting in RCD (Lan et al., 2025b), we construct a simulated task collection $\mathcal{D}_{\text{sim}} = \{(\mathbf{X}_i^{\text{sim}}, \mathbf{y}_i^{\text{sim}})\}_{i=1}^N$ covering diverse temporal structures and anomaly manifestations. Conceptually, the simulator induces a task distribution $P_s(\mathcal{T})$ over competence-relevant variations in temporal structures and anomaly mechanisms, rather than attempting to reproduce the real data distribution itself. This construction varies both normal temporal dynamics and anomaly mechanisms, since either can change the competence ordering among specialists. Each simulated task therefore serves as a *competence probe* for the specialist pool. The specialist pool is fixed across competence construction and target deployment and intentionally consists of lightweight detectors with heterogeneous inductive biases. This design makes routing primarily reflect which notion of normality and deviation is appropriate, rather than differences in model capacity. The complete pool and its corresponding inductive biases are detailed in Appendix B.1.3.

For each simulated task, every specialist M_j is evaluated under the Syn-RCD label-free fitting and scoring protocol and produces anomaly scores $a_{i,j}$. Since the simulated labels are known, its competence is evaluated as $c_{i,j}^{\text{sim}} = \Phi(\mathbf{a}_{i,j}, \mathbf{y}_i^{\text{sim}})$, where Φ is instantiated as VUS-PR and expressed on the $[0, 1]$ scale for competence construction. Thus, anomaly labels are used to evaluate specialist competence rather than to fit the specialists. The resulting profile $\mathbf{c}_i^{\text{sim}} = [c_{i,1}^{\text{sim}}, \dots, c_{i,K}^{\text{sim}}]^\top$ characterizes the relative suitability of different specialist biases for task i . Instead of reducing this profile to a single best-specialist label, we transform it into a soft competence target $\mathbf{q}_i \in \Delta^{K-1}$:

$$q_{i,j} = \frac{\exp(c_{i,j}^{\text{sim}}/\tau)}{\sum_{\ell=1}^K \exp(c_{i,\ell}^{\text{sim}}/\tau)}, \quad j = 1, \dots, K, \quad (4)$$

where $\tau > 0$ controls the sharpness of the target distribution. Soft targets preserve each task’s competence ordering and encode pairwise competence gaps through their log-ratios, including cases in which several specialists exhibit comparable competence. The resulting routing dataset $\mathcal{D}_{\text{route}} = \{(\mathbf{X}_i^{\text{sim}}, \mathbf{q}_i)\}_{i=1}^N$ provides scalable supervision for learning how specialist suitability varies with observable temporal context.

4.2 FOUNDATION-GUIDED COMPETENCE ROUTING

We combine a pretrained TSFM encoder with a trainable competence router. The encoder E_{TSFM} is obtained through pretraining under its native objective, independently of the specialist competence supervision used for routing.

During routing training, E_{TSFM} is frozen and used only as a feature extractor. Given a simulated series $\mathbf{X}_i^{\text{sim}}$, the final encoder layer produces contextualized token embeddings $\mathbf{H}_i = [\mathbf{h}_{i,1}, \dots, \mathbf{h}_{i,m}]^\top \in \mathbb{R}^{m \times D}$, where m is the number of temporal tokens and D is the representation dimension. Since competence routing is performed at the series level, we mean-pool all final-layer token embeddings to obtain

$$\mathbf{z}_i = \frac{1}{m} \sum_{j=1}^m \mathbf{h}_{i,j} \in \mathbb{R}^D.$$

The foundation model therefore provides a fixed temporal representation shared across specialists for inferring their suitability, rather than directly producing anomaly scores.

A lightweight router g_θ , implemented as an MLP, maps $\mathbf{z}_i$ to K specialist logits and predicts the competence distribution

$$\hat{\mathbf{q}}_i = \text{Softmax}(g_\theta(\mathbf{z}_i)) \in \Delta^{K-1}, \quad (5)$$

where $\hat{q}_{i,j}$ represents the predicted relative competence of specialist M_j . Predicting the full competence distribution preserves relative suitability across multiple specialists and aligns naturally with the soft targets defined in Section 4.1.

With E_{TSM} fixed, only the router parameters θ are optimized by matching $\hat{\mathbf{q}}_i$ to $\mathbf{q}_i$ using the Kullback–Leibler divergence:

$$\mathcal{L}_{\text{route}} = \frac{1}{N} \sum_{i=1}^N D_{\text{KL}}(\mathbf{q}_i \parallel \hat{\mathbf{q}}_i). \quad (6)$$

The resulting router predicts soft competence targets from fixed temporal representations.

4.3 OFFLINE TARGET ROUTING AND DETECTION

We study offline anomaly detection on an observed target series $\mathbf{X} = (\mathbf{X}^{\text{tr}}, \mathbf{X}^{\text{te}})$, where the within-series split designates a specialist-fitting prefix and an evaluation suffix. Both portions are available as unlabeled observations when routing is performed. The prefix specifies the observations used for specialist fitting; it does not delimit the input available to the router. Routing therefore conditions on the observed evaluation suffix as well as the prefix, while target anomaly labels are unavailable to routing, specialist fitting, and score generation.

The encoder is pretrained under its native objective, and the router is trained on $\mathcal{D}_{\text{route}}$. Both models are frozen before target deployment and process the complete unlabeled observed series only for forward inference; no target labels or encoder/router parameter updates are used. The router predicts the competence distribution $\hat{\mathbf{q}}(\mathbf{X})$, from which we select

$$\mathcal{S}_{\theta}(\mathbf{X}) = \{M_j : j \in \text{TopK}(\hat{\mathbf{q}}(\mathbf{X}), k)\}.$$

Each selected specialist $M_j \in \mathcal{S}_{\theta}(\mathbf{X})$ is fitted to $\mathbf{X}^{\text{tr}}$ without anomaly labels and produces a score sequence $\mathbf{a}_j(\mathbf{X}^{\text{te}})$ on the evaluation suffix. We standardize each sequence, average the selected outputs, and min–max rescale the fused scores:

$$\hat{\mathbf{a}}^{\text{te}} = \mathcal{M}\left(\frac{1}{k} \sum_{M_j \in \mathcal{S}_{\theta}(\mathbf{X})} \mathcal{N}(\mathbf{a}_j(\mathbf{X}^{\text{te}}))\right). \quad (7)$$

Here, $\mathcal{N}$ and $\mathcal{M}$ use statistics of the corresponding score sequences on the evaluation suffix, as specified in Appendix A.2. These operations produce continuous anomaly scores without target labels. For multivariate targets, the same observation protocol and routing–fitting–scoring procedure are applied independently to each variable, followed by the cross-variable aggregation described in that appendix. Algorithm 1 summarizes the complete procedure.

4.4 THEORETICAL ANALYSIS

We analyze Top- k selection at the deployment budget using the set competence in Eq. 1. Let P_s and P_r denote simulated and real task distributions, with $Z = E(\mathbf{X})$ denoting the fixed, mean-pooled TSMF representation. Throughout, $K \geq 2$, $1 \leq k \leq K$, $\tau > 0$, and $\mathbf{c}(\mathcal{T}) \in [0, 1]^K$. Write $\mathbf{q}(\mathcal{T}) = \text{Softmax}(\mathbf{c}(\mathcal{T})/\tau)$ and $\mathbf{p}_{\theta}(z) = \hat{\mathbf{q}}_{\theta}(z)$. The router is measurable and strictly positive, and selection uses a fixed measurable tie-breaking rule. Logarithms are natural, and $\text{osc}(\mathbf{v}) = \max_j v_j - \min_j v_j$.

Lemma 1 (Top- k Competence Regret from KL). Set $m_k = \min\{k, K - k\}$, $\alpha_k = m_k/k$, and $D_{\tau, K, k} = (k - 1)e^{1/\tau} + K - k + 1$. For $\text{osc}(\mathbf{v}) \leq 1$, $\mathbf{q} = \text{Softmax}(\mathbf{v}/\tau)$, and any strictly positive probability vector $\mathbf{p}$,

$$0 \leq T_k(\mathbf{v}) - V_k(\text{TopK}(\mathbf{p}, k), \mathbf{v}) \leq \min\left\{\alpha_k, C_{\tau, K, k} \sqrt{2D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p})}\right\}, \quad (8)$$

where

$$C_{\tau, K, k} = \min\left\{\frac{D_{\tau, K, k}}{k(1 - e^{-1/\tau})}, \frac{\sqrt{m_k D_{\tau, K, k}}}{k(1 - e^{-1/(2\tau)})}\right\}. \quad (9)$$

The proof pairs missed and incorrectly selected specialists and uses the lower bound $q_{(k)} \geq D_{\tau, K, k}^{-1}$ on the k -th largest target probability.

Define $\mu_d(z) = \mathbb{E}_{P_d}[\mathbf{c}(\mathcal{T}) \mid Z = z]$, $d \in \{s, r\}$, and $\mathbf{m}_s(z) = \mathbb{E}_{P_s}[\mathbf{q}(\mathcal{T}) \mid Z = z]$. Let $\mathcal{L}_s^*$ be the infimum of the population routing loss over all measurable simplex-valued predictors. The loss and its excess satisfy

$$\mathcal{L}_s(\theta) = \mathbb{E}_{P_s} D_{\text{KL}}(\mathbf{q}(\mathcal{T}) \parallel \mathbf{p}_\theta(Z)), \quad (10)$$

$$\mathcal{E}_s(\theta) := \mathcal{L}_s(\theta) - \mathcal{L}_s^* = \mathbb{E}_{P_s^Z} D_{\text{KL}}(\mathbf{m}_s(Z) \parallel \mathbf{p}_\theta(Z)). \quad (11)$$

The minimizer is $\mathbf{m}_s$, which need not equal $\text{Softmax}(\mu_s/\tau)$. We account for this difference through

$$b_{\tau,s}(z) = \text{osc}(\mu_s(z) - \tau \log \mathbf{m}_s(z)), \quad \beta_{\tau,r} = \mathbb{E}_{P_r^Z}[b_{\tau,s}(Z)]. \quad (12)$$

Assume $P_r^Z \ll P_s^Z$, with $dP_r^Z/dP_s^Z \leq \rho < \infty$, and $\|\mu_r(z) - \mu_s(z)\|_\infty \leq \delta$ for P_r^Z -almost every z .

The selected set has target competence $U_{\text{router},r}^{(k)} = \mathbb{E}_{P_r^Z}[V_k(\hat{S}_\theta^{(k)}(Z), \mu_r(Z))]$; a domain subscript specifies the distribution used to evaluate each utility.

Theorem 1 (Simulated-to-Real Top- k Routing Regret). Under the stated assumptions, every router with finite $\mathcal{L}_s(\theta)$ satisfies

$$U_{\text{cond},r}^{(k)}(Z) - U_{\text{router},r}^{(k)} \leq \min\left\{\alpha_k, C_{\tau,K,k} \sqrt{2\rho \mathcal{E}_s(\theta)} + \alpha_k(\beta_{\tau,r} + 2\delta)\right\}, \quad (13)$$

$$U_{\text{cond},r}^{(k)}(Z) - U_{\text{router},r}^{(k)} \leq \min\left\{\alpha_k, C_{\tau,K,k} \sqrt{2\rho \mathcal{L}_s(\theta)} + 2\alpha_k\delta\right\}. \quad (14)$$

Let $\epsilon_k(\theta)$ be the smaller right-hand side. Define the budget-specific representation gap and routing opportunity by

$$\Delta_{\text{rep}}^{(k)}(Z) = U_{\text{oracle},r}^{(k)} - U_{\text{cond},r}^{(k)}(Z), \quad \Gamma_{\text{route}}^{(k)}(Z) = U_{\text{cond},r}^{(k)}(Z) - U_{\text{fixed},r}^{(k)}. \quad (15)$$

Then

$$\begin{aligned} U_{\text{oracle},r}^{(k)} - U_{\text{router},r}^{(k)} &\leq \Delta_{\text{rep}}^{(k)}(Z) + \epsilon_k(\theta), \\ U_{\text{router},r}^{(k)} - U_{\text{fixed},r}^{(k)} &\geq \Gamma_{\text{route}}^{(k)}(Z) - \epsilon_k(\theta). \end{aligned} \quad (16)$$

The bound separates representation information, soft-target prediction, and competence transfer at a fixed budget. Proofs appear in Appendix A.1. Appendix A.1.5 relates recovery of the competence-optimal set to the fused prediction when the selection boundary is separated.

5 EXPERIMENTS

We evaluate TS-Router on 16 TSAD benchmarks to answer three questions: (1) *Overall effectiveness*: can TS-Router generalize across heterogeneous targets and outperform foundation-model-based and target-fitted TSAD methods? (2) *Routing effectiveness*: does competence-based routing improve over a globally fixed specialist, Best Top-3, and full-pool ensembling? (3) *Representation effectiveness*: do TSFM representations provide stronger signals for specialist competence estimation than conventional features, and are the benefits consistent across pretrained encoders? Table 1 addresses the first question across 16 benchmarks, while Table 2 examines the latter two on eleven univariate benchmarks to isolate routing and representation effects from variable-wise aggregation used for multivariate targets. Per-dataset results, implementation details, specialist complementarity, and representation analyses are provided in the appendix. Appendices B.2.4–B.2.6 cover budget, pool-size, and routing-head sensitivity; transfer tests show that TS-Router leads classical selectors under synthetic-to-real and leave-one-dataset-out settings.

5.1 EXPERIMENTAL SETUP

Datasets and Metrics. We evaluate 16 TSAD benchmarks, including eleven genuinely univariate collections (IOPS, MGAB, NAB, NEK, Power, SED, Stock, TODS, UCR, WSD, and YAHOO) and five multivariate datasets (MSL, PSM, SMAP, SMD, and SWaT). Following TSAD evaluation practice, we report VUS-PR, Affiliation-F1, F1_T, and Standard-F1, which capture complementary ranking-, event-, and point-level detection behavior. For routing analysis, NDCG@ k and Hit@ k measure agreement between the predicted specialist ranking and the VUS-PR-based ground-truth

Table 1: Overall performance across 16 real-world TSAD benchmarks. Score denotes the mean performance across datasets, and Rank denotes the average dataset-wise rank among the 12 compared methods, with ties assigned average ranks. Overall Rank averages ranks across all four metrics. #Top-1 and #Top-2 count first-place and top-two finishes across the 64 dataset–metric combinations. Methods follow different deployment paradigms without using target anomaly labels for model fitting. Best results are in **bold** and second-best results are underlined.

Method	VUS-PR		Aff.-F1		F1 _T		Std.-F1		Overall Rank	#Top-1	#Top-2
	Score↑	Rank↓	Score↑	Rank↓	Score↑	Rank↓	Score↑	Rank↓			
Foundation-guided specialist routing											
TS-Router	49.59	2.44	86.87	2.44	47.37	3.00	45.34	2.75	2.66	26	39
Direct zero-shot models											
TimeRCD	37.00	3.88	83.94	3.69	40.88	3.62	37.03	4.19	3.84	14	26
DADA	30.16	5.62	81.41	4.88	37.29	5.84	32.60	6.31	5.66	5	10
Chronos	26.66	7.09	80.67	5.31	32.54	6.91	28.52	7.12	6.61	2	8
MOMENT	28.57	6.81	74.39	6.94	25.66	7.09	22.31	7.50	7.09	0	3
TimesFM	28.59	6.44	70.45	8.12	32.08	7.44	28.71	7.44	7.36	1	8
Time-MoE	18.31	9.44	69.47	9.09	22.29	7.81	18.05	8.94	8.82	0	0
Target-fitted unsupervised models											
OmniAnomaly	30.73	4.75	75.40	6.12	34.44	4.38	31.60	4.44	4.92	10	17
USAD	29.44	5.50	68.67	7.94	30.83	5.28	30.00	4.75	5.87	5	13
TranAD	25.84	6.25	75.72	6.31	25.77	7.31	24.60	6.69	6.64	1	4
TFMAE	16.59	9.28	71.99	7.66	19.55	9.12	15.81	8.56	8.66	0	0
DCdetector	14.99	10.50	67.71	9.50	16.07	10.19	13.27	9.31	9.88	0	0

competence ranking. Detailed dataset statistics and preprocessing are provided in Appendix B.1.1. Threshold-dependent metrics use oracle thresholds selected from evaluation labels after score generation; these labels are never used for routing, specialist fitting, or anomaly-score generation.

Baselines and Deployment Protocols. We compare TS-Router with direct zero-shot models and target-fitted unsupervised TSAD methods. Neither detector fitting nor deployment-time specialist selection uses target anomaly labels, although deployment protocols differ. Direct zero-shot models perform inference without target-specific detector fitting, whereas target-fitted methods and the specialists selected by TS-Router adapt unsupervisedly to each target series. Target anomaly labels are used only for evaluation, including hindsight reference strategies. Full baseline descriptions and implementation details are provided in Appendix B.1.5. For the controlled routing and representation comparisons, all applicable selectors and feature extractors receive the same unlabeled target observations. Specialist fitting data, evaluation intervals, and score-normalization and fusion rules are matched wherever applicable.

TS-Router Configuration. TS-Router uses a fixed pool of eleven lightweight specialists with heterogeneous inductive biases. By default, we instantiate E_{TSFM} with the frozen TimeRCD encoder, and construct routing supervision from specialists’ VUS-PR competence on labeled Syn-RCD tasks. The same learned router is evaluated under all four detection metrics. At deployment, the router selects the Top-3 specialists before fitting; only these specialists are fitted unsupervisedly to the target series and their normalized scores are mean-fused. Representation ablations replace the default encoder with conventional feature representations or frozen Chronos and MOMENT encoders, training a router for each representation under the same competence supervision and deployment protocol. Unless otherwise stated, all experiments use the default configuration. Specialist definitions, hyperparameters, and encoder-specific implementation details are provided in the appendix.

5.2 EXPERIMENTAL RESULTS

Overall Performance. Table 1 shows that TS-Router achieves the highest mean detection scores and the best average ranks under all four metrics across the 16 benchmarks. It ranks first in 26 of the 64 dataset–metric combinations and among the top two in 39, indicating benefits across multiple targets

and metrics. These results hold against both direct zero-shot models and target-fitted unsupervised detectors, although their deployment protocols differ. In particular, TS-Router reuses the pretrained encoder underlying TimeRCD but uses its representation to estimate specialist competence, followed by target-specific fitting of selected detectors. This comparison demonstrates the effectiveness of the complete routing-and-adaptation pipeline; the controlled analyses below examine how specialist selection and temporal representation contribute to its performance. Appendix B.2.7 shows that TS-Router leads classical selectors under synthetic-to-real and leave-one-dataset-out transfer, while remaining competitive when they receive labeled in-distribution support.

Table 2: Routing effectiveness and representation ablations on eleven univariate benchmarks. Best Fixed denotes the hindsight-best single specialist shared across targets for each evaluation metric, whereas Oracle denotes hindsight per-target individual-specialist selection; these are single-specialist detection references, not Top-3 fusion oracles. Best Top-3 always fuses the hindsight-best set of three specialists shared across targets. Routing metrics use $k=1$ for Best Fixed and Oracle and $k=3$ otherwise; Router Top-1 shares the TS-Router ranking, so its routing metrics are not repeated. Representation variants differ only in the router representation, with the same router architecture, specialist pool, supervision, and Top-3 normalized mean fusion. The dashed line separates conventional feature representations from frozen TSFM encoders (Chronos, MOMENT, TimeRCD); TimeRCD corresponds to the default TS-Router configuration.

Variant	Detection performance				Routing quality	
	VUS-PR $\uparrow$	Aff.-F1 $\uparrow$	F1 $_T$ $\uparrow$	Std.-F1 $\uparrow$	NDCG@ k $\uparrow$	Hit@ k $\uparrow$
<i>Selection and fusion strategy</i>						
Best Fixed	39.68	84.34	41.01	36.43	0.607	0.207
Full Ensemble	49.54	85.55	46.22	43.48	–	–
Best Fixed Top-3	47.44	86.56	47.56	43.42	0.625	0.532
Router Top-1	49.00	86.80	43.47	41.41	–	–
TS-Router Top-3	56.36	88.11	49.62	47.73	0.782	0.632
Oracle	68.05	94.29	66.59	64.59	1.000	1.000
<i>Router representation (Top-3 routing)</i>						
Catch22	49.96	84.65	42.28	40.13	0.674	0.426
TSFresh	49.01	88.16	47.13	42.86	0.656	0.570
BasicStats	45.48	84.32	44.60	41.31	0.584	0.436
MiniRocket	51.31	85.91	44.63	43.53	0.660	0.496
Chronos	53.27	87.25	45.93	42.79	0.698	0.494
MOMENT	53.89	88.50	49.74	46.37	0.721	0.629
TimeRCD	56.36	88.11	49.62	47.73	0.782	0.632

Effectiveness of Competence Routing. The upper block of Table 2 examines specialist selection and its downstream use in detection. TS-Router outperforms the hindsight-best fixed individual specialist and full-pool ensembling under all four detection metrics. Best Top-3 always fuses the hindsight-best set of three specialists shared across targets, yet remains below TS-Router Top-3 on both detection and routing metrics, supporting target-adaptive selection over that globally fixed set. The NDCG and Hit scores assess agreement with individual specialist competence before fusion. Top-3 improves over Router Top-1 under all four detection metrics, and TS-Router’s advantage over full-pool ensembling supports selective rather than indiscriminate fusion. These comparisons evaluate the detection effect of combining the selected score sequences, which is separate from the fixed-budget set-competence analysis.

Role of the Foundation Representation. The lower block of Table 2 examines the representation supporting this competence estimation, holding the router architecture, specialist pool, supervision, and fusion procedure fixed. All three frozen TSFM encoders achieve higher VUS-PR and NDCG than the conventional feature representations. TimeRCD provides the highest NDCG and Hit scores, together with the best VUS-PR and Standard-F1, while MOMENT achieves the highest Affiliation-F1 and F1 $_T$. The results with Chronos and MOMENT show that effective foundation-guided routing is not confined to the default TimeRCD encoder. Differences across metrics also

indicate that stronger ranking quality need not yield uniformly better detection scores. Taken together, the two blocks support the representation-to-competence-to-detection pathway: pretrained representations inform specialist suitability, competence-based routing identifies useful subsets, and selective fusion improves detection. Consistent with Section 3, these findings support the generalist-representation, specialist-detection perspective: transferable temporal knowledge guides which criteria to trust, while selected specialists perform anomaly detection.

6 CONCLUSION

We introduced TS-Router, a generalist-representation, specialist-detection framework that reconsiders the role of foundation models in time-series anomaly detection. TS-Router uses transferable temporal representations to estimate specialist competence and select suitable detectors, with labeled simulation providing soft routing supervision without requiring large-scale real anomaly annotations. Our analysis bounds Top- k set-competence regret under explicit transfer assumptions, while experiments evaluate the complete representation-to-competence-to-detection pathway: foundation representations inform specialist suitability, and selective score fusion improves detection. Results with multiple frozen TSFM encoders further show that this role is not confined to a single pretrained model. As with other simulation-based approaches, competence transfer is shaped by how well the simulated task distribution covers variations relevant to unseen targets. For multivariate series, we focus on variable-wise routing to isolate the representation-to-competence-to-detection pathway; incorporating cross-variable interactions is a natural extension.

AI USE STATEMENT

Generative AI tools were used during the research and manuscript preparation for brainstorming and discussing alternative formulations of the research problem, providing feedback on experimental design and interpretation of empirical results, assisting with drafting, restructuring, and polishing parts of the manuscript, and serving as a checking and discussion aid for the theoretical component. For mathematical arguments and proofs, generative AI was used primarily to inspect derivations, identify possible inconsistencies or unclear assumptions, and help assess whether proof steps and stated conclusions were logically aligned, rather than to replace the authors’ own mathematical reasoning. The final TS-Router formulation, assumptions, proofs, experimental procedures, reported results, and conclusions were independently examined and determined by the authors. Generative AI was not used to discover or retrieve related literature or to generate the simulated datasets used in the experiments. All AI-assisted suggestions were critically reviewed before inclusion, and the authors take full responsibility for the final content of this work, including its text, mathematical claims, experimental results, and conclusions.

ETHICS STATEMENT

This work studies time-series anomaly detection using public real-world benchmark datasets and simulated Syn-RCD tasks. It does not involve human subjects, personally identifiable information, or interventions in real-world systems. The method is intended for research on anomaly detection; deployment in safety-critical domains requires domain-specific validation and appropriate human oversight.

REPRODUCIBILITY STATEMENT

We provide detailed descriptions of TS-Router’s competence-construction, router-training, and target-deployment procedures in Section 4, Algorithm 1, and Appendix B. Dataset descriptions, pre-processing, specialist definitions and hyperparameters, baseline configurations, evaluation metrics, and implementation settings are reported in Appendix B. Additional per-dataset results, representation ablations, specialist complementarity analyses, Top- k and specialist-pool sensitivity, routing-head ablations, and algorithm-selection transfer experiments are also provided in Appendix B. Complete assumptions and proofs of the theoretical results are given in Appendix A. The supplementary materials provide the Syn-RCD simulation specifications, competence-target construction details, and implementation resources needed to reproduce the experiments. The code is available at <https://anonymous.4open.science/r/TS-Router-D8FF>.

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A ADDITIONAL METHODOLOGICAL DETAILS

A.1 THEORETICAL SUPPLEMENT

This section proves the budget- k regret and transfer results, establishes the associated utility hierarchy, and relates specialist-set recovery to score fusion. Throughout, $K \geq 2$, $1 \leq k \leq K$, $\tau > 0$, and $\mathbf{c}(\mathcal{T}) \in [0, 1]^K$. The encoder, pooling operation, specialist pool, and label-free fitting and scoring protocol are fixed. Task and representation spaces are standard Borel spaces, so regular conditional distributions exist. All conditional statements hold almost surely under the relevant representation distribution. The router $\mathbf{p}_\theta(z) = \hat{\mathbf{q}}_\theta(z)$ is measurable and strictly positive. Every Top- k operation uses the same fixed measurable tie-breaking rule and returns a set of indices in $\mathcal{I}_k$. We use the notation V_k , T_k , m_k , α_k , and $C_{\tau,K,k}$ from Sections 3 and 4.4.

A.1.1 PROOF OF LEMMA 1

Equal-budget set differences. For any $S, S' \in \mathcal{I}_k$, the sets $A = S \setminus S'$ and $B = S' \setminus S$ have the same cardinality $r \leq \min\{k, K - k\} = m_k$, since $|S| = |S'| = k$. Canceling the common members $S \cap S'$ and pairing the remaining indices by any bijection $\pi : A \rightarrow B$ gives, for every $\mathbf{u} \in \mathbb{R}^K$,

$$V_k(S, \mathbf{u}) - V_k(S', \mathbf{u}) = \frac{1}{k} \sum_{i \in A} (u_i - u_{\pi(i)}).$$

Each summand lies in $[-\text{osc}(\mathbf{u}), \text{osc}(\mathbf{u})]$, so

$$|V_k(S, \mathbf{u}) - V_k(S', \mathbf{u})| \leq \frac{r}{k} \text{osc}(\mathbf{u}) \leq \alpha_k \text{osc}(\mathbf{u}). \quad (17)$$

In particular, every selection regret for a vector of range at most one is bounded by α_k . If $k = K$, then $\mathcal{I}_K$ is a singleton, $m_K = 0$, and both the regret and the second term in $C_{\tau,K,K}$ vanish, so $C_{\tau,K,K} = 0$. Henceforth take $k < K$.

Probability mass at the selection boundary. Let $v_{(1)} \geq \dots \geq v_{(K)}$ denote the ordered coordinates of $\mathbf{v}$, and let $q_{(k)}$ be the k -th largest coordinate of $\mathbf{q} = \text{Softmax}(\mathbf{v}/\tau)$. Softmax is invariant under subtracting $v_{(k)}$, so

$$q_{(k)} = \frac{e^{v_{(k)}/\tau}}{\sum_{\ell=1}^K e^{v_{(\ell)}/\tau}} = \left(\sum_{\ell=1}^K e^{(v_{(\ell)} - v_{(k)})/\tau} \right)^{-1}.$$

Since $\text{osc}(\mathbf{v}) \leq 1$, one has $v_{(\ell)} - v_{(k)} \leq 1$ for $\ell < k$ and $v_{(\ell)} - v_{(k)} \leq 0$ for $\ell > k$. Thus the first $k-1$ summands are at most $e^{1/\tau}$, the remaining $K-k+1$ summands (including $\ell = k$) are at most one, and

$$q_{(k)} \geq \frac{1}{(k-1)e^{1/\tau} + K - k + 1} = \frac{1}{D_{\tau,K,k}}. \quad (18)$$

Let $S^* = \text{TopK}(\mathbf{v}, k)$ and $\hat{S} = \text{TopK}(\mathbf{p}, k)$. If the two sets agree, the regret is zero. Otherwise, write $A = S^* \setminus \hat{S}$, $B = \hat{S} \setminus S^*$, and $|A| = |B| = r \geq 1$. Choose any bijection $\pi : A \rightarrow B$. For $i \in A$, put $d_i = v_i - v_{\pi(i)}$. Because S^* maximises $V_k(\cdot, \mathbf{v})$, every $i \in S^*$ and $j \notin S^*$ satisfy $v_i \geq v_j$, hence $d_i \geq 0$. The range bound $\text{osc}(\mathbf{v}) \leq 1$ gives $d_i \leq 1$. Softmax preserves the order of $\mathbf{v}$ and the same tie-breaking rule, so $S^* = \text{TopK}(\mathbf{q}, k)$ and therefore $q_i \geq q_{(k)} \geq D_{\tau,K,k}^{-1}$. The softmax ratios yield $q_{\pi(i)}/q_i = e^{-d_i/\tau}$. Because $\hat{S}$ maximises $V_k(\cdot, \mathbf{p})$, one has $p_{\pi(i)} \geq p_i$. Collecting these relations,

$$0 \leq d_i \leq 1, \quad q_i \geq D_{\tau,K,k}^{-1}, \quad q_{\pi(i)} = q_i e^{-d_i/\tau}, \quad p_{\pi(i)} \geq p_i. \quad (19)$$

Canceling common members of S^* and $\hat{S}$ gives

$$\mathsf{T}_k(\mathbf{v}) - V_k(\hat{S}, \mathbf{v}) = \frac{1}{k} \sum_{i \in A} d_i.$$

Probability differences. The map $x \mapsto 1 - e^{-x/\tau}$ is concave, so on $[0, 1]$ it lies above the chord joining 0 and 1: $1 - e^{-d_i/\tau} \geq d_i(1 - e^{-1/\tau})$. Combined with $q_i - q_{\pi(i)} = q_i(1 - e^{-d_i/\tau})$ and

$$q_i \geq D_{\tau,K,k}^{-1},$$

$$q_i - q_{\pi(i)} \geq \frac{1 - e^{-1/\tau}}{D_{\tau,K,k}} d_i.$$

For the comparison with $\mathbf{p}$, write

$$q_i - q_{\pi(i)} = (q_i - p_i) + (p_i - p_{\pi(i)}) + (p_{\pi(i)} - q_{\pi(i)}).$$

The middle term is nonpositive by Eq.(19), so

$$q_i - q_{\pi(i)} \leq |q_i - p_i| + |q_{\pi(i)} - p_{\pi(i)}|.$$

The pairs $(i, \pi(i))$ are disjoint. Summing therefore yields

$$\begin{aligned} \frac{1 - e^{-1/\tau}}{D_{\tau,K,k}} \sum_{i \in A} d_i &\leq \sum_{i \in A} (q_i - q_{\pi(i)}) \\ &\leq \sum_{j \in A \cup B} |q_j - p_j| \leq \|\mathbf{q} - \mathbf{p}\|_1. \end{aligned} \quad (20)$$

Pinsker's inequality states that $\|\mathbf{q} - \mathbf{p}\|_1 \leq \sqrt{2D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p})}$. To verify the constant, when $\mathbf{q} \neq \mathbf{p}$, let $I = \{j : q_j \geq p_j\}$, $a = \sum_{j \in I} q_j$, and $b = \sum_{j \in I} p_j$. Then $a \geq b$ and $\|\mathbf{q} - \mathbf{p}\|_1 = 2(a - b)$. Since $\mathbf{p}$ is strictly positive, $b \in (0, 1)$. The log-sum inequality gives

$$D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p}) \geq \text{kl}(a \parallel b), \quad \text{kl}(a \parallel b) = a \log \frac{a}{b} + (1 - a) \log \frac{1 - a}{1 - b}.$$

For fixed $b \in (0, 1)$, this binary divergence and its first derivative vanish at $a = b$, while its second derivative is $1/[a(1 - a)] \geq 4$. Taylor expansion with remainder therefore yields $\text{kl}(a \parallel b) \geq 2(a - b)^2 = \frac{1}{2} \|\mathbf{q} - \mathbf{p}\|_1^2$. The case $\mathbf{q} = \mathbf{p}$ is immediate. Applying this inequality to Eq.(20) yields

$$\frac{1}{k} \sum_{i \in A} d_i \leq \frac{D_{\tau,K,k}}{k(1 - e^{-1/\tau})} \sqrt{2D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p})}. \quad (21)$$

Square-root probability differences. The same chord bound at temperature 2τ and $q_i \geq D_{\tau,K,k}^{-1}$ give

$$\sqrt{q_i} - \sqrt{q_{\pi(i)}} = \sqrt{q_i}(1 - e^{-d_i/(2\tau)}) \geq \frac{1 - e^{-1/(2\tau)}}{\sqrt{D_{\tau,K,k}}} d_i.$$

Because $p_{\pi(i)} \geq p_i$,

$$\sqrt{q_i} - \sqrt{q_{\pi(i)}} \leq |\sqrt{q_i} - \sqrt{p_i}| + |\sqrt{q_{\pi(i)}} - \sqrt{p_{\pi(i)}}|.$$

Summing over the r disjoint pairs and applying Cauchy-Schwarz on the $2r$ coordinates in $A \cup B$ yields

$$\frac{1 - e^{-1/(2\tau)}}{\sqrt{D_{\tau,K,k}}} \sum_{i \in A} d_i \leq \sqrt{2r} H(\mathbf{q}, \mathbf{p}), \quad H^2(\mathbf{q}, \mathbf{p}) = \sum_{j=1}^K (\sqrt{q_j} - \sqrt{p_j})^2. \quad (22)$$

Let $a(\mathbf{q}, \mathbf{p}) = \sum_j \sqrt{q_j p_j}$. Then $H^2(\mathbf{q}, \mathbf{p}) = 2 - 2a(\mathbf{q}, \mathbf{p})$. Jensen's inequality for the concave logarithm and $-\log x \geq 1 - x$ imply

$$D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p}) = -2 \sum_j q_j \log \sqrt{\frac{p_j}{q_j}} \geq -2 \log a(\mathbf{q}, \mathbf{p}) \geq 2(1 - a(\mathbf{q}, \mathbf{p})) = H^2(\mathbf{q}, \mathbf{p}). \quad (23)$$

Combining Eqs.(22) and (23) and using $H \leq \sqrt{D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p})}$ together with $r \leq m_k$ gives

$$\frac{1}{k} \sum_{i \in A} d_i \leq \frac{\sqrt{m_k D_{\tau,K,k}}}{k(1 - e^{-1/(2\tau)})} \sqrt{2D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p})}. \quad (24)$$

Taking the smaller of Eqs.(21) and (24), together with the bound α_k from Eq.(17), proves Lemma 1. $\square$

A.1.2 PROOF OF THEOREM 1

Population target of the KL objective. For fixed z , abbreviate $\mathbf{q} = \mathbf{q}(\mathcal{T})$, $\mathbf{m} = \mathbf{m}_s(z)$, and $\mathbf{p} = \mathbf{p}_\theta(z)$. Since $\mathbf{p}$ and $\mathbf{m}$ are functions of z and $\mathbb{E}_{P_s}[q_j \mid Z = z] = m_j$, splitting $\log(q_j/p_j) = \log(q_j/m_j) + \log(m_j/p_j)$ gives

$$\mathbb{E}_{P_s} \left[\sum_{j=1}^K q_j \log \frac{q_j}{p_j} \mid Z = z \right] = \mathbb{E}_{P_s} \left[\sum_{j=1}^K q_j \log \frac{q_j}{m_j} \mid Z = z \right] + \sum_{j=1}^K m_j \log \frac{m_j}{p_j},$$

which is

$$\mathbb{E}_{P_s}[D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p}) \mid Z = z] = \mathbb{E}_{P_s}[D_{\text{KL}}(\mathbf{q} \parallel \mathbf{m}) \mid Z = z] + D_{\text{KL}}(\mathbf{m} \parallel \mathbf{p}). \quad (25)$$

Because $\mathbf{c}(\mathcal{T}) \in [0, 1]^K$,

$$q_j(\mathcal{T}) = \frac{e^{c_j(\mathcal{T})/\tau}}{\sum_{\ell=1}^K e^{c_\ell(\mathcal{T})/\tau}} \geq K^{-1} e^{-1/\tau} > 0,$$

and the same lower bound passes to $\mathbf{m}_s$ by conditional expectation. The target-dependent logarithms are therefore bounded, and finite $\mathcal{L}_s(\theta)$ justifies integration. The second term in Eq.(25) is minimized at the measurable predictor $\mathbf{p}(z) = \mathbf{m}_s(z)$. Therefore

$$\mathcal{L}_s^* = \mathbb{E}_{P_s} D_{\text{KL}}(\mathbf{q}(\mathcal{T}) \parallel \mathbf{m}_s(Z)), \quad (26)$$

$$\mathcal{E}_s(\theta) = \mathbb{E}_{P_s^Z} D_{\text{KL}}(\mathbf{m}_s(Z) \parallel \mathbf{p}_\theta(Z)). \quad (27)$$

This infimum is over all measurable simplex-valued predictors, not only the chosen router architecture.

Range of the conditional score vector. Write $\mathbf{h}_s(z) = \tau \log \mathbf{m}_s(z)$, with the logarithm taken coordinatewise. Since $|c_i - c_j| \leq 1$, each task satisfies

$$e^{-1/\tau} q_j(\mathcal{T}) \leq q_i(\mathcal{T}) \leq e^{1/\tau} q_j(\mathcal{T}).$$

These are linear inequalities in $\mathbf{q}(\mathcal{T})$, so they pass to the conditional means: $e^{-1/\tau} m_{s,j}(z) \leq m_{s,i}(z) \leq e^{1/\tau} m_{s,j}(z)$. Taking logarithms yields $|\tau \log(m_{s,i}(z)/m_{s,j}(z))| \leq 1$. Consequently,

$$\text{osc}(\mathbf{h}_s(z)) \leq 1, \quad \text{Softmax}(\mathbf{h}_s(z)/\tau) = \mathbf{m}_s(z). \quad (28)$$

Lemma 1 thus applies to $\mathbf{h}_s(z)$, even though the conditional mean soft target need not be the softmax of the conditional mean competence. Also, $0 \leq b_{\tau,s}(z) \leq \text{osc}(\boldsymbol{\mu}_s(z)) + \text{osc}(\mathbf{h}_s(z)) \leq 2$, so the distortion is integrable.

Conditional Top- k regret. For $d \in \{s, r\}$, define

$$R_{d,k}(z) = \mathsf{T}_k(\boldsymbol{\mu}_d(z)) - V_k(\widehat{S}_\theta^{(k)}(z), \boldsymbol{\mu}_d(z)). \quad (29)$$

Fix z , set $\widehat{S} = \widehat{S}_\theta^{(k)}(z)$, and let $S_s^* = \text{TopK}(\boldsymbol{\mu}_s(z), k)$. Write $\mathbf{e}_s = \boldsymbol{\mu}_s - \mathbf{h}_s$, so that $b_{\tau,s}(z) = \text{osc}(\mathbf{e}_s(z))$ by Eq.(12). Then

$$R_{s,k}(z) = V_k(S_s^*, \boldsymbol{\mu}_s(z)) - V_k(\widehat{S}, \boldsymbol{\mu}_s(z)) = V_k(S_s^*, \mathbf{h}_s(z)) - V_k(\widehat{S}, \mathbf{h}_s(z)) + V_k(S_s^*, \mathbf{e}_s(z)) - V_k(\widehat{S}, \mathbf{e}_s(z)).$$

The first difference is at most $\mathsf{T}_k(\mathbf{h}_s(z)) - V_k(\widehat{S}, \mathbf{h}_s(z))$. The second is at most $\alpha_k b_{\tau,s}(z)$ by Eq.(17). Lemma 1 applied to $\mathbf{h}_s(z)$, using Eq.(28), therefore gives

$$R_{s,k}(z) \leq C_{\tau,K,k} \sqrt{2D_{\text{KL}}(\mathbf{m}_s(z) \parallel \mathbf{p}_\theta(z))} + \alpha_k b_{\tau,s}(z). \quad (30)$$

There is also a bound directly in terms of the total routing loss. Let

$$\eta_s(z) = \mathbb{E}_{P_s}[D_{\text{KL}}(\mathbf{q}(\mathcal{T}) \parallel \mathbf{p}_\theta(z)) \mid Z = z].$$

The function T_k is convex, being a maximum of finitely many linear functions, and $\widehat{S}$ is fixed conditional on $Z = z$. Jensen's inequality and linearity of $V_k(\widehat{S}, \cdot)$ therefore yield

$$R_{s,k}(z) = \mathsf{T}_k(\boldsymbol{\mu}_s(z)) - V_k(\widehat{S}, \boldsymbol{\mu}_s(z)) \leq \mathbb{E}_{P_s} [\mathsf{T}_k(\mathbf{c}(\mathcal{T})) - V_k(\widehat{S}, \mathbf{c}(\mathcal{T})) \mid Z = z].$$

Lemma 1 applies task-wise because $\text{osc}(\mathbf{c}(\mathcal{T})) \leq 1$. Concavity of the square root then gives

$$\begin{aligned} R_{s,k}(z) &\leq C_{\tau,K,k} \mathbb{E}_{P_s} \left[\sqrt{2D_{\text{KL}}(\mathbf{q}(\mathcal{T}) \parallel \mathbf{p}_\theta(z))} \mid Z = z \right] \\ &\leq C_{\tau,K,k} \sqrt{2\eta_s(z)}. \end{aligned} \quad (31)$$

The tower property gives $\mathbb{E}_{P_s^Z} \eta_s(Z) = \mathcal{L}_s(\theta)$.

Conditional competence shift. Write $\mathbf{d}(z) = \boldsymbol{\mu}_r(z) - \boldsymbol{\mu}_s(z)$ and $S_r^* = \text{TopK}(\boldsymbol{\mu}_r(z), k)$. Then

$$\begin{aligned} R_{r,k}(z) &= V_k(S_r^*, \boldsymbol{\mu}_s(z) + \mathbf{d}(z)) - V_k(\hat{S}, \boldsymbol{\mu}_s(z) + \mathbf{d}(z)) \\ &= (V_k(S_r^*, \boldsymbol{\mu}_s(z)) - V_k(\hat{S}, \boldsymbol{\mu}_s(z))) + (V_k(S_r^*, \mathbf{d}(z)) - V_k(\hat{S}, \mathbf{d}(z))). \end{aligned}$$

The first difference is at most $R_{s,k}(z)$. The second is at most $\alpha_k \text{osc}(\mathbf{d}(z))$ by Eq.(17). The assumption $\|\boldsymbol{\mu}_r - \boldsymbol{\mu}_s\|_\infty \leq \delta$ gives $\text{osc}(\mathbf{d}(z)) \leq 2\delta$, hence

$$R_{r,k}(z) \leq R_{s,k}(z) + \alpha_k \text{osc}(\mathbf{d}(z)) \leq R_{s,k}(z) + 2\alpha_k \delta. \quad (32)$$

Transfer across representation distributions. Let $w = dP_r^Z / dP_s^Z$. Then $0 \leq w \leq \rho$, so $w^2 \leq \rho w$. For any nonnegative measurable f with finite source expectation, the Cauchy–Schwarz inequality $\mathbb{E}[w\sqrt{f}] \leq \sqrt{\mathbb{E}[w^2] \mathbb{E}[f]}$ and $\mathbb{E}_{P_s^Z} w = 1$ imply

$$\mathbb{E}_{P_r^Z} \sqrt{f(Z)} = \mathbb{E}_{P_s^Z} [w(Z) \sqrt{f(Z)}] \leq \sqrt{\mathbb{E}_{P_s^Z} [w(Z)^2] \mathbb{E}_{P_s^Z} [f(Z)]} \leq \sqrt{\rho \mathbb{E}_{P_s^Z} [f(Z)]}. \quad (33)$$

Absolute continuity transfers source-almost-sure conditional bounds to P_r^Z . Taking target expectations in Eq.(32) and using Eq.(30) gives

$$\begin{aligned} \mathbb{E}_{P_r^Z} R_{r,k}(Z) &\leq C_{\tau,K,k} \mathbb{E}_{P_r^Z} \sqrt{2D_{\text{KL}}(\mathbf{m}_s(Z) \parallel \mathbf{p}_\theta(Z))} + \alpha_k \beta_{\tau,r} + 2\alpha_k \delta \\ &\leq C_{\tau,K,k} \sqrt{2\rho \mathcal{E}_s(\theta)} + \alpha_k (\beta_{\tau,r} + 2\delta). \end{aligned} \quad (34)$$

Using Eq.(31) instead, together with $\mathbb{E}_{P_s^Z} \eta_s(Z) = \mathcal{L}_s(\theta)$, gives

$$\mathbb{E}_{P_r^Z} R_{r,k}(Z) \leq C_{\tau,K,k} \sqrt{2\rho \mathcal{L}_s(\theta)} + 2\alpha_k \delta. \quad (35)$$

By definition, $\mathbb{E}_{P_r^Z} R_{r,k}(Z) = U_{\text{cond},r}^{(k)}(Z) - U_{\text{router},r}^{(k)}$. Each coordinate of $\boldsymbol{\mu}_r(z)$ lies in $[0, 1]$, so $\text{osc}(\boldsymbol{\mu}_r(z)) \leq 1$ and Eq.(17) yields $0 \leq R_{r,k}(z) \leq \alpha_k$. Combining this trivial bound with Eqs.(34) and (35) proves both inequalities in Theorem 1, including the case $k = K$, where $\alpha_K = C_{\tau,K,K} = 0$. $\square$

A.1.3 CONDITIONAL SOFTMAX DISTORTION

The distortion in Eq.(12) measures differences between conditional competence gaps and log-ratios of conditional mean soft targets. For fixed P_s , E , and τ , it is independent of the router parameters.

Coordinate-wise Jensen gaps. For $j \in [K]$, define

$$J_j(z) = \log \mathbb{E}_{P_s} [q_j(\mathcal{T}) \mid Z = z] - \mathbb{E}_{P_s} [\log q_j(\mathcal{T}) \mid Z = z] \geq 0. \quad (36)$$

Let $F_\tau(\mathbf{c}) = \tau \log \sum_{\ell=1}^K e^{c_\ell/\tau}$. Then $\tau \log q_j = c_j - F_\tau(\mathbf{c})$, so

$$\mathbb{E}_{P_s} [\tau \log q_j(\mathcal{T}) \mid Z = z] = \mu_{s,j}(z) - \mathbb{E}_{P_s} [F_\tau(\mathbf{c}(\mathcal{T})) \mid Z = z].$$

The definition of J_j rearranges to $\tau \log m_{s,j}(z) = \mathbb{E}_{P_s} [\tau \log q_j(\mathcal{T}) \mid Z = z] + \tau J_j(z)$, hence

$$h_{s,j}(z) = \mu_{s,j}(z) - \mathbb{E}_{P_s} [F_\tau(\mathbf{c}(\mathcal{T})) \mid Z = z] + \tau J_j(z).$$

The conditional log-normalizer does not depend on j , so

$$\boldsymbol{\mu}_s(z) - \mathbf{h}_s(z) = \mathbb{E}_{P_s} [F_\tau(\mathbf{c}(\mathcal{T})) \mid Z = z] \mathbf{1} - \tau \mathbf{J}(z)$$

and therefore

$$b_{\tau,s}(z) = \text{osc}(\boldsymbol{\mu}_s(z) - \mathbf{h}_s(z)) = \tau \text{osc}(\mathbf{J}(z)). \quad (37)$$

A common additive gap across specialists does not change $\text{osc}(\mathbf{J})$.

A sufficient condition for zero distortion. Suppose that, under P_s ,

$$\mathbf{c}(\mathcal{T}) = \mathbf{v}(Z) + a(\mathcal{T})\mathbf{1}, \quad (38)$$

where $\mathbf{v}$ is measurable and a is integrable. This means that Z determines all pairwise competence differences, while overall competence levels may vary through a common task-dependent offset. Softmax is invariant under such offsets, so $\mathbf{q}(\mathcal{T}) = \text{Softmax}(\mathbf{v}(Z)/\tau)$ is conditionally deterministic. Hence $\mathbf{J}(z) = \mathbf{0}$, $b_{\tau,s}(z) = 0$, and absolute continuity gives $\beta_{\tau,r} = 0$. Conversely, if all differences $c_j - c_1$ are measurable functions of Z , Eq.(38) holds with $v_1 = 0$, $v_j(Z) = c_j - c_1$, and $a = c_1$.

For fixed K, k, τ, ρ , a sequence of routers with $\mathcal{E}_s(\theta_n) \rightarrow 0$, together with $\beta_{\tau,r} = 0$ and $\delta = 0$, therefore satisfies

$$U_{\text{cond},r}^{(k)}(Z) - U_{\text{router},r}^{(k)}(\theta_n) \rightarrow 0. \quad (39)$$

When distortion is nonzero, it remains explicit in the bound.

A quantitative bound under conditional concentration. Suppose that, under P_s conditional on $Z = z$, each $Y_j = \tau \log q_j(\mathcal{T})$ lies in an interval of length at most $r(z)$, where r is measurable and $\mathbb{E}_{P_r^Z}[r(Z)^2] < \infty$. Then

$$b_{\tau,s}(z) \leq \frac{r(z)^2}{8\tau}, \quad \beta_{\tau,r} \leq \frac{\mathbb{E}_{P_r^Z}[r(Z)^2]}{8\tau}. \quad (40)$$

To prove this, fix z, j , write $Y = Y_j$, $r = r(z)$, and set $G(t) = \log \mathbb{E}_{P_s}[e^{tY} \mid Z = z]$. Then $G(0) = 0$ and $G'(0) = \mathbb{E}_{P_s}[Y \mid Z = z]$. Under exponential tilting, $G''(t)$ is the variance of a random variable supported on an interval of length r . If m denotes the midpoint of that interval, $\mathbb{E}[(Y - m)^2] \leq (r/2)^2 = r^2/4$, and the variance is at most this second moment, so $G''(t) \leq r^2/4$. The integral remainder

$$G(t) - tG'(0) = \int_0^t (t-u)G''(u) du \leq \frac{r^2 t^2}{8}, \quad t \geq 0,$$

together with $J_j(z) = G(1/\tau) - (1/\tau)G'(0)$, yields $0 \leq J_j(z) \leq r(z)^2/(8\tau^2)$. Therefore $\tau \text{osc}(\mathbf{J}(z)) \leq \tau \max_j J_j(z) \leq r(z)^2/(8\tau)$, proving Eq.(40) after target averaging.

A.1.4 UTILITY HIERARCHY AND REPRESENTATION INFORMATION

Fixed, conditional, and hindsight selection. Under any task distribution, write $\boldsymbol{\mu}(Z) = \mathbb{E}[\mathbf{c}(\mathcal{T}) \mid Z]$. Linearity of $V_k(S, \cdot)$ and the tower property give $\mathbb{E}[V_k(S, \mathbf{c}(\mathcal{T}))] = \mathbb{E}_Z[V_k(S, \boldsymbol{\mu}(Z))]$ for every fixed $S \in \mathcal{I}_k$. Maximising over S and using $\mathbf{T}_k(\boldsymbol{\mu}(Z)) = \max_S V_k(S, \boldsymbol{\mu}(Z))$ yields

$$U_{\text{fixed}}^{(k)} = \max_{S \in \mathcal{I}_k} \mathbb{E}_Z[V_k(S, \boldsymbol{\mu}(Z))] \leq \mathbb{E}_Z[\mathbf{T}_k(\boldsymbol{\mu}(Z))] = U_{\text{cond}}^{(k)}(Z), \quad (41)$$

$$U_{\text{cond}}^{(k)}(Z) = \mathbb{E}_Z[\mathbf{T}_k(\mathbb{E}[\mathbf{c}(\mathcal{T}) \mid Z])] \leq \mathbb{E}_Z[\mathbf{T}_k(\mathbf{c}(\mathcal{T})) \mid Z] = U_{\text{oracle}}^{(k)}, \quad (42)$$

where the second comparison is Jensen's inequality for the convex map $\mathbf{T}_k$. The conditional optimum is attained by the measurable set $\text{TopK}(\boldsymbol{\mu}(Z), k)$, establishing Eq.(3).

If $Z_0 = h(Z_1)$ for a measurable h , then $\mathbb{E}[\mathbf{c} \mid Z_0] = \mathbb{E}[\mathbb{E}[\mathbf{c} \mid Z_1] \mid Z_0]$. Conditional Jensen for $\mathbf{T}_k$ gives

$$\mathbf{T}_k(\boldsymbol{\mu}(Z_0)) \leq \mathbb{E}[\mathbf{T}_k(\boldsymbol{\mu}(Z_1)) \mid Z_0],$$

and taking expectations yields

$$U_{\text{cond}}^{(k)}(Z_0) \leq \mathbb{E}[\mathbb{E}[\mathbf{T}_k(\mathbb{E}[\mathbf{c} \mid Z_1]) \mid Z_0]] = U_{\text{cond}}^{(k)}(Z_1). \quad (43)$$

This monotonicity concerns the best attainable conditional competence at the same budget, not the performance of every trained router.

Representation and learned selection. Because $\widehat{S}_\theta^{(k)}(Z)$ is Z -measurable, the tower property gives

$$U_{\text{router},r}^{(k)} = \mathbb{E}_{P_r}[V_k(\widehat{S}_\theta^{(k)}(Z), \mathbf{c}(\mathcal{T}))] = \mathbb{E}_{P_r^Z}[V_k(\widehat{S}_\theta^{(k)}(Z), \boldsymbol{\mu}_r(Z))].$$

Adding and subtracting $U_{\text{cond},r}^{(k)}(Z)$ therefore yields

$$\begin{aligned} U_{\text{oracle},r}^{(k)} - U_{\text{router},r}^{(k)} &= \Delta_{\text{rep}}^{(k)}(Z) + \mathbb{E}_{P_r^Z} R_{r,k}(Z), \\ U_{\text{router},r}^{(k)} - U_{\text{fixed},r}^{(k)} &= \Gamma_{\text{route}}^{(k)}(Z) - \mathbb{E}_{P_r^Z} R_{r,k}(Z). \end{aligned}$$

Theorem 1 supplies $\mathbb{E}_{P_r^Z} R_{r,k}(Z) \leq \epsilon_k(\theta)$, which is Eq.(16).

Coverage and relative competence stability. Theorem 1 separates changes in the distribution of representations from changes in competence at a given representation. Its change-of-measure argument also admits an integrated form. Suppose $P_r^Z \ll P_s^Z$, write $w = dP_r^Z/dP_s^Z$, and define

$$\kappa_Z = \mathbb{E}_{P_s^Z} [w(Z)^2] < \infty, \quad \Delta_{\text{rel}} = \mathbb{E}_{P_r^Z} \text{osc}(\boldsymbol{\mu}_r(Z) - \boldsymbol{\mu}_s(Z)).$$

The first bound in Eq.(33) with $\mathbb{E}_{P_s^Z} [w^2] = \kappa_Z$ in place of ρ , together with $\text{osc}(\boldsymbol{\mu}_r - \boldsymbol{\mu}_s)$ in place of 2δ in Eq.(32) and $R_{r,k} \leq \alpha_k$, yields

$$\mathbb{E}_{P_r^Z} R_{r,k}(Z) \leq \min \left\{ \alpha_k, C_{\tau,K,k} \sqrt{2\kappa_Z \mathcal{E}_s(\theta)} + \alpha_k(\beta_{\tau,r} + \Delta_{\text{rel}}) \right\}, \quad (44)$$

$$\mathbb{E}_{P_r^Z} R_{r,k}(Z) \leq \min \left\{ \alpha_k, C_{\tau,K,k} \sqrt{2\kappa_Z \mathcal{L}_s(\theta)} + \alpha_k \Delta_{\text{rel}} \right\}. \quad (45)$$

The assumptions of Theorem 1 give $\kappa_Z \leq \rho$ and $\Delta_{\text{rel}} \leq 2\delta$. If $\boldsymbol{\mu}_r(z) - \boldsymbol{\mu}_s(z)$ is constant across coordinates, then $\Delta_{\text{rel}} = 0$ and the two Top- k sets of $\boldsymbol{\mu}_r(z)$ and $\boldsymbol{\mu}_s(z)$ coincide.

A.1.5 FROM SET RECOVERY TO SCORE FUSION

Theorem 1 bounds the competence of the selected set. We next compare the resulting fused scores with those obtained from the conditionally competence-optimal set $S_r^{*,(k)}(z) = \text{TopK}(\boldsymbol{\mu}_r(z), k)$. Let Ψ be a measurable detection metric on the $[0, 1]$ scale. For $S \in \mathcal{I}_k$, define

$$F_{\mathcal{T}}(S) = \Psi \left(\mathcal{M} \left(\frac{1}{k} \sum_{j \in S} \mathcal{N}(\mathbf{a}_j(\mathbf{X})) \right), \mathbf{y} \right) \in [0, 1], \quad (46)$$

where scores and labels are restricted to the evaluation interval, and $\mathcal{N}$ and $\mathcal{M}$ are the standardization and min-max transformations in Appendix A.2. Under a fixed fitting and normalization protocol, identical selected sets yield identical fused scores.

Corollary 1 (Boundary Separation and Fused-Utility Agreement). For $1 \leq k < K$, write $\mu_{r,(1)}(z) \geq \dots \geq \mu_{r,(K)}(z)$ and define the boundary gap

$$\gamma_k(z) = \mu_{r,(k)}(z) - \mu_{r,(k+1)}(z). \quad (47)$$

Under the assumptions of Theorem 1, for every $t > 0$,

$$\begin{aligned} \mathbb{E}_{P_r} \left| F_{\mathcal{T}}(S_r^{*,(k)}(Z)) - F_{\mathcal{T}}(\widehat{S}_{\theta}^{(k)}(Z)) \right| \\ \leq \Pr_{P_r^Z} \left(\widehat{S}_{\theta}^{(k)}(Z) \neq S_r^{*,(k)}(Z) \right) \leq \min \left\{ 1, \Pr_{P_r^Z}(\gamma_k(Z) \leq t) + \frac{k\epsilon_k(\theta)}{t} \right\}. \end{aligned} \quad (48)$$

For $k = K$, the two sets and their fused outputs coincide identically.

Proof. On the event $\gamma_k(z) > t$, the set $S_r^{*,(k)}(z)$ is the unique collection of the k largest coordinates of $\boldsymbol{\mu}_r(z)$, and every member has conditional competence more than t above every excluded specialist. If $r \geq 1$ members are replaced, pairing each missed index i with an incorrectly selected index $\pi(i)$ gives

$$\mu_{r,i}(z) - \mu_{r,\pi(i)}(z) \geq \gamma_k(z) > t,$$

and therefore

$$R_{r,k}(z) \geq \frac{r\gamma_k(z)}{k} > \frac{t}{k}.$$

Splitting the mismatch event according to whether $\gamma_k(z) \leq t$ then yields

$$\mathbf{1}\{\widehat{S}_{\theta}^{(k)}(z) \neq S_r^{*,(k)}(z)\} \leq \mathbf{1}\{\gamma_k(z) \leq t\} + \frac{kR_{r,k}(z)}{t}.$$

Taking expectations and applying Theorem 1 bounds the mismatch probability by $\Pr_{P_r^Z}(\gamma_k(Z) \leq t) + k\epsilon_k(\theta)/t$. Since $F_{\mathcal{T}} \in [0, 1]$, its absolute difference is at most one on a mismatch and is zero when the sets agree, giving the first inequality. The probability is at most one, which proves Eq.(48). $\square$

The mismatch bound depends on the gap γ_k between the k -th and $(k+1)$ -th specialists. If $\epsilon_k(\theta_n) \rightarrow 0$, then for every $t > 0$,

$$\limsup_{n \rightarrow \infty} \Pr_{P_r^Z}(\hat{S}_{\theta_n}^{(k)}(Z) \neq S_r^{*,(k)}(Z)) \leq \Pr_{P_r^Z}(\gamma_k(Z) \leq t).$$

Sending $t \downarrow 0$ and using $\Pr_{P_r^Z}(\gamma_k(Z) = 0) = 0$ yields set recovery and convergence of the expected absolute fused-utility difference.

A.1.6 SCOPE OF THE GUARANTEE AND EXPERIMENTAL INTERPRETATION

Mean individual competence V_k and the fused detection utility $F_{\mathcal{T}}$ in Eq.(46) are different objectives. Theorem 1 compares selectors at the same budget through V_k . Corollary 1 bounds the fused-score difference that arises when the selected set differs from $S_r^{*,(k)}$; it does not rank detection performance across values of k . The empirical budget sweep in Appendix B.2.4 evaluates that additional fusion tradeoff. The reported NDCG and Hit values assess predicted specialist rankings, while the detection metrics assess the complete fitting-and-fusion pipeline. The core analysis concerns a single routed series and applies to each variable-wise selection in the multivariate protocol.

A.2 OVERALL TRAINING AND DEPLOYMENT PROCEDURE

Algorithm 1 summarizes the complete competence-construction, router-training, and target-deployment procedure of TS-Router. The procedure assumes access to a pretrained temporal encoder E_{TSFM} , which is kept frozen throughout competence learning. In our main experiments, E_{TSFM} is instantiated with the pretrained TimeRCD encoder. The labeled simulated tasks are used only to evaluate the relative competence of unsupervised specialists and construct routing targets; they do not provide supervision for fitting the specialists themselves. At deployment, routing is performed before detector fitting, so only the selected specialists are fitted on the training prefix of the unlabeled target series.

For the main experiments, we set $k = 3$. Importantly, the complete specialist pool is evaluated when constructing synthetic competence supervision, whereas real-target deployment first predicts specialist suitability and then fits only the selected Top- k specialists. Thus, target anomaly labels and deployment-time evaluation of the complete candidate pool are both unnecessary for routing.

Score Normalization. Because heterogeneous specialists may produce anomaly scores on different scales, each anomaly-score sequence is independently standardized before fusion. For a score sequence $\mathbf{a} = (a_1, \dots, a_L)$, we define

$$\mathcal{N}(\mathbf{a})_t = \frac{a_t - \mu(\mathbf{a})}{\sigma(\mathbf{a})}, \quad t = 1, \dots, L, \quad (49)$$

where $\mu(\mathbf{a})$ and $\sigma(\mathbf{a})$ denote the mean and standard deviation of the score sequence within the current segment. The standardized scores of the selected specialists are then combined by an unweighted mean. The fused sequence $\bar{\mathbf{a}}$ is subsequently min-max rescaled to $[0, 1]$:

$$\mathcal{M}(\bar{\mathbf{a}})_t = \frac{\bar{a}_t - \min(\bar{\mathbf{a}})}{\max(\bar{\mathbf{a}}) - \min(\bar{\mathbf{a}})}, \quad t = 1, \dots, L, \quad (50)$$

so that

$$\hat{\mathbf{a}}(\mathbf{X}) = \mathcal{M}\left(\frac{1}{k} \sum_{M_j \in \mathcal{S}_\theta(\mathbf{X})} \mathcal{N}(\mathbf{a}_j(\mathbf{X}))\right). \quad (51)$$

For multivariate targets, the same per-specialist standardization is applied independently for each variable. Neither normalization nor fusion uses target anomaly labels.

Multivariate Targets. For a multivariate target $\mathbf{X} \in \mathbb{R}^{L \times V}$, we apply the same routing and detection procedure independently to each variable $\mathbf{X}^{(v)}$, $v = 1, \dots, V$. Each variable therefore has its own predicted competence profile and selected specialist subset,

$$\mathcal{S}_\theta(\mathbf{X}^{(v)}) = \{M_j : j \in \text{TopK}(\hat{\mathbf{q}}(\mathbf{X}^{(v)}), k)\}. \quad (52)$$

The selected specialists are fitted independently to $\mathbf{X}^{(v)}$. Their standardized scores are mean-fused and then min-max rescaled to produce the variable-wise score

$$\hat{\mathbf{a}}^{(v)} = \mathcal{M}\left(\frac{1}{k} \sum_{M_j \in \mathcal{S}_\theta(\mathbf{X}^{(v)})} \mathcal{N}(\mathbf{a}_j(\mathbf{X}^{(v)}))\right). \quad (53)$$

Before cross-variable aggregation, each variable-wise score sequence is again standardized with $\mathcal{N}$. The standardized variable scores are averaged and then min-max rescaled:

$$\hat{\mathbf{a}}_{\text{multi}} = \mathcal{M}\left(\frac{1}{V} \sum_{v=1}^V \mathcal{N}(\hat{\mathbf{a}}^{(v)})\right). \quad (54)$$

This two-stage normalization prevents differences in either specialist-score scale or variable-score scale from dominating the final prediction. The variable-wise treatment preserves the same routing mechanism across univariate and multivariate benchmarks without introducing a dimension-specific multivariate router.

Algorithm 1 Training and deployment of TS-Router.

Require: Simulated tasks $\mathcal{D}_{\text{sim}} = \{(\mathbf{X}_i^{\text{sim}}, \mathbf{y}_i^{\text{sim}})\}_{i=1}^N$; pretrained encoder E_{TSFM} ; specialist pool $\mathcal{M} = \{M_1, \dots, M_K\}$; competence utility Φ ; temperature τ ; number of selected specialists k

Ensure: Trained router g_θ and target anomaly scores

- 1: **Stage I: Synthetic competence construction**
 - 2: Freeze E_{TSFM}
 - 3: **for** each simulated task $(\mathbf{X}_i^{\text{sim}}, \mathbf{y}_i^{\text{sim}}) \in \mathcal{D}_{\text{sim}}$ **do**
 - 4: Split $\mathbf{X}_i^{\text{sim}}$ into a fitting prefix $\mathbf{X}_i^{\text{sim}, \text{tr}}$ and an evaluation suffix $\mathbf{X}_i^{\text{sim}, \text{te}}$ without using anomaly labels
 - 5: Let $\mathbf{y}_i^{\text{sim}, \text{te}}$ denote the labels restricted to the evaluation suffix
 - 6: **for** each specialist $M_j \in \mathcal{M}$ **do**
 - 7: Fit M_j to $\mathbf{X}_i^{\text{sim}, \text{tr}}$ without anomaly labels
 - 8: Obtain anomaly scores $\mathbf{a}_{i,j}^{\text{sim}} = M_j(\mathbf{X}_i^{\text{sim}, \text{te}})$
 - 9: Compute competence $c_{i,j}^{\text{sim}} = \Phi(\mathbf{a}_{i,j}^{\text{sim}}, \mathbf{y}_i^{\text{sim}, \text{te}})$
 - 10: **end for**
 - 11: Construct the soft competence target
$$q_{i,j} = \frac{\exp(c_{i,j}^{\text{sim}}/\tau)}{\sum_{\ell=1}^K \exp(c_{i,\ell}^{\text{sim}}/\tau)}, \quad j = 1, \dots, K$$
 - 12: Extract final-layer token representations from the complete unlabeled series $\mathbf{H}_i = E_{\text{TSFM}}(\mathbf{X}_i^{\text{sim}})$
 - 13: Mean-pool the token representations
$$\mathbf{z}_i = \frac{1}{m} \sum_{t=1}^m \mathbf{h}_{i,t}$$
 - 14: **end for**
 - 15: **Stage II: Router learning**
 - 16: **while** the router has not converged **do**
 - 17: Predict competence distributions $\hat{\mathbf{q}}_i = \text{Softmax}(g_\theta(\mathbf{z}_i))$
 - 18: Update θ by minimizing
$$\mathcal{L}_{\text{route}} = \frac{1}{N} \sum_{i=1}^N D_{\text{KL}}(\mathbf{q}_i \parallel \hat{\mathbf{q}}_i)$$
 - 19: **end while**
 - 20: **Stage III: Label-free target deployment**
 - 21: Receive an unlabeled target series $\mathbf{X}$
 - 22: Extract and mean-pool its temporal representation
$$\mathbf{z} = \text{MeanPool}(E_{\text{TSFM}}(\mathbf{X}))$$
 - 23: Predict relative specialist competence $\hat{\mathbf{q}}(\mathbf{X}) = \text{Softmax}(g_\theta(\mathbf{z}))$
 - 24: Select specialists before fitting
$$\mathcal{S}_\theta(\mathbf{X}) = \{M_j : j \in \text{TopK}(\hat{\mathbf{q}}(\mathbf{X}), k)\}$$
 - 25: Split $\mathbf{X}$ into a training prefix $\mathbf{X}^{\text{tr}}$ and a test suffix $\mathbf{X}^{\text{te}}$
 - 26: **for** each $M_j \in \mathcal{S}_\theta(\mathbf{X})$ **do**
 - 27: Fit M_j to $\mathbf{X}^{\text{tr}}$ without anomaly labels
 - 28: Obtain anomaly scores $\mathbf{a}_j(\mathbf{X}^{\text{te}})$
 - 29: Normalize scores as $\tilde{\mathbf{a}}_j(\mathbf{X}^{\text{te}}) = \mathcal{N}(\mathbf{a}_j(\mathbf{X}^{\text{te}}))$
 - 30: **end for**
 - 31: Fuse the selected specialists
$$\bar{\mathbf{a}}(\mathbf{X}^{\text{te}}) = \frac{1}{k} \sum_{M_j \in \mathcal{S}_\theta(\mathbf{X})} \tilde{\mathbf{a}}_j(\mathbf{X}^{\text{te}})$$
 - 32: Min-max rescale the fused scores
$$\hat{\mathbf{a}}(\mathbf{X}^{\text{te}}) = \mathcal{M}(\bar{\mathbf{a}}(\mathbf{X}^{\text{te}}))$$
 - 33: **return** $\hat{\mathbf{a}}(\mathbf{X}^{\text{te}})$
-

Table 3: Statistics of the univariate benchmark datasets.

Name	Domain	#TS	Avg. Length	AR (%)
UCR	Misc.	228	67818.7	0.6
NAB	Mixed	28	5099.7	10.6
YAHOO	Web	259	1560.2	0.6
IOPS	Operations	17	72792.3	1.3
MGAB	Synthetic	9	97777.8	0.2
SED	Medical	3	23332.3	4.1
Stock	Finance	20	15000.0	9.4
TODS	Synthetic	15	5000.0	6.3
NEK	Web	9	1073.0	8.0
Power	Power Grid	1	35040.0	8.5
WSD	Web	111	17444.5	0.6

Table 4: Statistics of the multivariate benchmark datasets.

Name	Domain	#TS	Avg. Length	AR (%)
MSL	Space	16	3119.4	5.1
PSM	Sensor	1	217624.0	11.2
SMAP	Space	27	7855.9	2.9
SMD	Server	22	25466.4	3.8
SWaT	ICS	2	207457.5	12.7

B ADDITIONAL EXPERIMENTAL DETAILS

B.1 EXPERIMENTAL SETUP

B.1.1 BENCHMARK DATASETS

Our evaluation follows the TSB-AD benchmark and uses a collection of genuinely univariate and multivariate TSAD datasets. A substantial portion of the univariate track in TSB-AD is obtained by decomposing multivariate datasets into individual variables. To avoid evaluating closely related series in both the univariate and multivariate settings, we exclude these decomposed multivariate series from the univariate evaluation and retain only genuinely univariate collections. This results in eleven univariate benchmarks: IOPS, MGAB, NAB, NEK, Power, SED, Stock, TODS, UCR, WSD, and YAHOO. We additionally evaluate on five standard multivariate benchmarks: MSL, PSM, SMAP, SMD, and SWaT. Together, these datasets cover heterogeneous domains, temporal dynamics, and anomaly characteristics.

The eleven genuinely univariate collections are also used for the controlled routing and representation analyses in the main text. This avoids introducing the additional variable-wise aggregation involved in multivariate detection when isolating the effects of specialist selection and representation choice. Tables 3 and 4 summarize the dataset statistics, where #TS denotes the number of time series, Avg. Length denotes the average number of observations per series, and AR denotes the anomaly ratio.

B.1.2 BASELINE METHODS AND DEPLOYMENT PROTOCOLS

We compare TS-Router with two groups of TSAD baselines that represent different ways of using transferable or target-specific temporal knowledge. All compared methods are evaluated without access to target anomaly labels. However, their deployment protocols differ: direct zero-shot models apply a pretrained model without target-specific detector fitting, whereas target-fitted unsupervised methods learn or adapt their detection mechanism from each unlabeled target series. TS-Router belongs to neither category directly: it uses a fixed pretrained representation and routing policy to select specialists, after which only the selected specialists are fitted unsupervisedly to the tar-

get series. The comparison therefore evaluates different deployment paradigms under the common constraint of no target anomaly labels.

Direct Zero-Shot Models. This group contains pretrained time-series models that can be applied to unseen target series without target-specific detector fitting.

- **TimeRCD** is an anomaly-specific time-series foundation model pretrained with labeled synthetic data for zero-shot anomaly detection. In the baseline setting, its pretrained detector is directly applied to each target without target-specific fitting.
- **DADA** is a pretrained general anomaly detector designed for cross-dataset deployment. Following the original experimental setting, we use an input window length of 100.
- **Chronos** formulates time-series forecasting through a language-modeling-style generative framework. We derive anomaly evidence from its pretrained predictive behavior and use an input window length of 100.
- **MOMENT** is a general-purpose time-series foundation model trained with patch-based temporal representation learning. We use an input window length of 64.
- **TimesFM** is a decoder-only pretrained time-series model designed primarily for zero-shot forecasting. We use an input window length of 96.
- **Time-MoE** is a decoder-only time-series foundation model with a sparse mixture-of-experts architecture. We use an input window length of 96.

Target-Fitted Unsupervised Models. This group contains representative deep TSAD methods that are fitted separately to each unlabeled target dataset or series. They therefore exploit target-specific normal temporal structure but do not use target anomaly labels.

- **OmniAnomaly** uses a stochastic recurrent architecture with variational latent variables to model normal multivariate temporal dynamics and detects anomalies through deviations from the learned normal model. We use an input window length of 100.
- **USAD** employs adversarially trained autoencoders for reconstruction-based anomaly detection. We use an input window length of 100.
- **TranAD** uses a transformer-based reconstruction framework to model temporal dependencies and scores anomalies from reconstruction discrepancies. We use an input window length of 10.
- **TFMAE** is a masked-autoencoder-based TSAD method that learns temporal representations by reconstructing masked time-series segments.
- **DCdetector** uses dual-attention contrastive representation learning to model temporal and channel-wise dependencies for anomaly detection.

For external baselines, we follow the configurations used in the original implementations or the benchmark protocol wherever applicable. Direct zero-shot models are not refitted on the target data, whereas target-fitted unsupervised models are trained independently on each target without anomaly labels. This distinction is preserved throughout all reported comparisons.

B.1.3 SPECIALIST POOL

TS-Router uses a fixed pool of eleven lightweight unsupervised anomaly detectors with heterogeneous inductive biases. Each detector is treated as a *specialist* because it embodies a distinct criterion for modeling normality and measuring deviation, rather than corresponding to a predefined anomaly type. The pool spans subspace reconstruction, marginal density, neighborhood structure, one-class boundaries, isolation, prototype distance, and spectral or temporal structure. This diversity provides complementary views of abnormality across heterogeneous time series.

The use of lightweight specialists is intentional. By limiting differences in model capacity, routing primarily reflects which inductive bias is appropriate for a target series rather than which candidate has the largest model capacity. Their low fitting cost also supports the route-before-fit deployment of TS-Router, where only the selected specialists are fitted to the unlabeled target series.

Table 5 summarizes the specialist pool and its primary detection bias.

Table 5: Specialists and their primary anomaly-detection inductive biases.

Specialist	Primary inductive bias
PCA	Low-dimensional subspace reconstruction; anomalies produce large reconstruction residuals.
Sub-PCA	Windowed PCA on subsequences; anomalies produce large reconstruction residuals relative to local temporal structure.
Sub-HBOS	Marginal density modeling through histogram-based statistics; low-density observations receive larger anomaly scores.
POLY	Smooth temporal structure modeled by polynomial fitting; deviations from the fitted temporal trend indicate anomalies.
Sub-KNN	Neighborhood-distance structure; observations far from their nearest neighbors are considered anomalous.
Sub-LOF	Relative local density; anomalies exhibit substantially lower local density than their surrounding neighborhoods.
KMeansAD	Prototype-based structure; anomaly scores are determined by distance from learned cluster centers.
OCSVM	One-class decision boundary; anomalies lie outside the region representing the dominant normal data distribution.
Sub-OCSVM	One-class boundary modeling on windowed subsequences.
Sub-IForest	Isolation structure; anomalous observations require fewer random partitions to isolate.
SR	Spectral residual structure; anomalies correspond to salient deviations from regular spectral-temporal patterns.

The same specialist pool is used when constructing synthetic competence supervision and when deploying TS-Router on real targets. During competence construction, all eleven specialists are fitted independently to each labeled simulated task without using anomaly labels for fitting, and their VUS-PR performance defines the competence profile. At deployment, the router predicts the relative suitability of the same specialists before fitting and selects only the Top-3 candidates for target-specific unsupervised adaptation. Detailed detector hyperparameters are provided in Appendix B.1.5.

B.1.4 EVALUATION METRICS

We evaluate anomaly detection performance using four complementary metrics: VUS-PR, Affiliation-F1, $F1_T$, and Standard-F1. Time-series anomalies often occur as contiguous intervals rather than isolated points, so relying on a single point-wise metric may not adequately reflect temporal overlap, event localization, and tolerance to boundary shifts. The four metrics therefore provide complementary point-wise, event-level, and range-aware views of detection quality.

Standard-F1. Standard-F1 is the conventional point-wise F1 score computed from true positives, false positives, and false negatives. It measures the harmonic mean of point-wise precision and recall and provides a basic measure of anomaly localization accuracy.

$F1_T$. $F1_T$ is a time-series range-wise F1 measure that accounts for temporal overlap and proximity between predicted and ground-truth anomaly intervals. Compared with purely point-wise evaluation, it better reflects cases where anomalous intervals are partially detected or predictions are slightly shifted relative to the annotated regions.

Affiliation-F1. Affiliation-F1 evaluates predicted anomaly events through an affiliation-based matching mechanism that accounts for their temporal proximity to ground-truth events. It is less sensitive to small boundary misalignments than point-wise F1 and therefore provides an event-oriented view of detection quality.

Oracle-threshold evaluation. Standard-F1, $F1_T$, and Affiliation-F1 are computed from binary predicted labels obtained by thresholding continuous anomaly scores. The choice of threshold rule changes the meaning of the resulting F1 values: a threshold selected from evaluation labels is not interchangeable with a label-free, pre-specified, or train-time threshold. Throughout this paper, we

apply a uniform *oracle-threshold* protocol to all three metrics and all compared methods. After a method has produced evaluation scores, we search candidate thresholds on those scores and retain the threshold that maximizes the corresponding F1 metric against the evaluation labels. This search is performed independently for each method, each target series, and each F1 metric. Evaluation labels are used only to choose the threshold after scores have been produced; they are not used for encoder training, router training, specialist fitting, or routing. The reported F1 numbers should therefore be read as oracle-threshold evaluation rather than as operational performance when labels are unavailable at threshold selection. This convention does not use labels for model training and does not invalidate the threshold-independent VUS-PR results, but it does limit the strength of deployment claims that would require choosing a threshold without evaluation labels.

VUS-PR. VUS-PR is a threshold-independent, range-aware metric based on the precision–recall curve. It evaluates detection performance over different temporal tolerance ranges and integrates the resulting precision–recall behavior into a volume-under-the-surface score. This makes it suitable for settings where anomaly intervals may exhibit temporal lag or boundary uncertainty. We additionally use VUS-PR as the competence utility Φ when constructing routing supervision, so the router is trained from a single consistent competence criterion while its downstream detection performance is evaluated under all four metrics.

For routing analysis, we further report $\text{NDCG}@k$ and $\text{Hit}@k$, which compare the specialist ranking predicted by the router with the ground-truth competence ranking obtained from specialists’ VUS-PR performance on each target task. $\text{NDCG}@k$ evaluates the quality of the predicted ordering while giving greater weight to highly ranked specialists. $\text{Hit}@k$ equals 1 when the predicted top- k set contains a specialist that attains the highest VUS-PR on the target task, and equals 0 otherwise. Higher values indicate stronger agreement with the VUS-PR-based ground-truth specialist ranking.

For Best Fixed and Oracle, routing quality is evaluated with $k = 1$, whereas Top-3 selection strategies are evaluated with $k = 3$. Router Top-1 and TS-Router Top-3 use the same predicted ranking. Table 2 reports that ranking’s $\text{NDCG}@3$ and $\text{Hit}@3$ once, in the TS-Router Top-3 row; the Router Top-1 row reports its detection performance only. Sharing a ranking does not imply equality of ranking metrics evaluated at different cutoffs.

B.1.5 IMPLEMENTATION DETAILS

This section records the implementation settings used throughout the main experiments. Unless otherwise stated, TS-Router uses the same frozen encoder, router, specialist pool, Top-3 selection, and fusion rule.

Encoder and Competence Supervision. We instantiate E_{TSFM} with the publicly released TimeRCD encoder and keep it frozen during competence learning and deployment. Given a series, we z-score normalize the input, extract the final-layer token representations, and mean-pool them to a 256-dimensional vector $\mathbf{z}$. Routing supervision is constructed from 850,000 labeled SynRCD tasks. Each simulated series is partitioned into a specialist-fitting prefix and an evaluation suffix. Every specialist is fitted on the prefix without anomaly labels and evaluated by VUS-PR on the suffix, following the same prefix-fitting and suffix-scoring protocol as at deployment. The resulting competence profile is converted into a soft target with temperature $\tau = 0.1$ as in Eq. (4). The encoder receives the complete unlabeled simulated series, including both portions, while anomaly labels are used only to evaluate specialist competence and construct routing targets.

Router Architecture and Training. The router g_θ is an MLP with hidden widths 1024, 512, and 256 and ReLU activations. It maps $\mathbf{z} \in \mathbb{R}^{256}$ to K specialist logits, which are converted into $\hat{\mathbf{q}}$ by a softmax. Only the router parameters are optimized. We minimize the average Kullback–Leibler objective in Eq. (6) with Adam, learning rate 10^{-4} , weight decay 10^{-5} , batch size 512, and 30 training epochs. The random seed is 42.

Specialist Hyperparameters. The eleven specialists use a fixed hyperparameter configuration across synthetic competence construction and real-target deployment. Sliding-window lengths are obtained from the TSB-AD period rank (periodicity). Table 6 lists the settings used in all reported experiments.

Deployment Protocol. Following TSB-AD, each target series is split into a training prefix and a test suffix. The series is z-score normalized before specialist fitting. Under the offline observation

Table 6: Specialist hyperparameters used for competence construction and target deployment.

Specialist	Hyperparameters
PCA	Period rank 1; all principal components.
Sub-PCA	Period rank 1; all principal components.
Sub-HBOS	Period rank 1; 10 histogram bins.
POLY	Period rank 1; polynomial degree 4.
Sub-KNN	Period rank 2; 50 neighbors.
Sub-LOF	Period rank 2; 30 neighbors.
KMeansAD	Period rank 2; 10 clusters.
OCSVM	Period rank 1; RBF kernel; $\nu = 0.5$.
Sub-OCSVM	Period rank 2; RBF kernel; $\nu = 0.5$.
Sub-IForest	Period rank 1; 150 trees.
SR	Period rank 1.

protocol, the frozen encoder and router receive the complete unlabeled target series, including both portions, and select the Top-3 specialists before fitting. Only these specialists are then fitted on the training prefix without anomaly labels and produce anomaly scores on the test suffix. Their scores are standardized, mean-fused, and min-max rescaled as in Appendix A.2. All detection metrics are computed on the test suffix. Multivariate targets use the same variable-wise routing and two-stage aggregation described there. The encoder and router remain frozen throughout target deployment; target observations are used for forward inference and specialist fitting and scoring, without updating the encoder or router parameters.

Baseline Configurations. Direct zero-shot models use the input window lengths stated in Appendix B.1.2. For TimeRCD we additionally use window length 5000 and batch size 64. Target-fitted unsupervised models use the same window lengths as in Appendix B.1.2, with learning rates 0.002 for OmniAnomaly, 0.001 for USAD, and 0.0001 for TranAD. Remaining baseline settings follow the original implementations or the TSB-AD protocol.

B.2 EXTENDED EXPERIMENTS

This section provides additional empirical results complementing the aggregate comparisons in the main text. We first report complete per-dataset detection results, followed by controlled analyses of the router representation, specialist complementarity, Top- k selection, specialist-pool size, and algorithm-selection transfer.

B.2.1 PER-DATASET PERFORMANCE

Table 7 reports the complete per-dataset results corresponding to the aggregate comparison in Table 1. The results reveal substantial variation in the relative strengths of different methods across datasets and evaluation metrics. No competing detector consistently dominates across heterogeneous targets, including both pretrained zero-shot models and target-fitted unsupervised methods. This variation is consistent with the motivation of TS-Router: different temporal contexts can favor different anomaly-detection criteria.

These per-dataset results complement the average-rank analysis in the main text.

B.2.2 DETAILED REPRESENTATION ABLATION

We further examine how the representation supplied to the router affects specialist competence estimation. Table 8 reports the complete per-dataset results corresponding to the representation ablation in Table 2. All variants use the same MLP router, eleven-specialist pool, competence supervision, Top-3 selection, and normalized mean fusion; only the input representation is changed. This controlled setting isolates whether the representation itself provides useful information for predicting specialist suitability.

TimeRCD achieves the best average performance on VUS-PR and Standard-F1. MOMENT is strongest on Affiliation-F1 and $F1_T$, remaining close to TimeRCD on the latter. These differences

Table 7: Per-dataset performance across 16 real-world TSAD benchmarks. Best results are in **bold** and second-best results are underlined.

Metric	Model	Univariate datasets										Multivariate datasets					
		IOPS	MGAB	NAB	NEK	Power	SED	Stock	TODS	UCR	WSD	YAHOO	MSL	PSM	SMAP	SMD	SWaT
VUS-PR	TS-Router	41.49	43.09	51.97	76.92	22.40	65.42	73.18	71.65	39.60	56.91	77.32	20.86	19.55	38.12	52.24	42.75
	TimeRCD	20.23	1.05	24.32	27.88	21.25	80.75	77.28	93.46	23.09	21.77	84.41	20.45	18.69	22.68	37.03	17.58
	DADA	24.97	0.57	24.73	46.85	10.61	6.42	99.51	64.83	2.94	33.42	70.74	12.74	17.17	20.02	25.98	21.13
	Chronos	19.00	0.60	23.76	31.80	10.95	8.65	97.49	70.66	6.56	18.81	83.54	8.25	14.61	5.18	10.22	16.44
	MOMENT	37.35	0.56	45.38	67.74	10.50	4.31	76.97	56.45	6.17	55.26	30.81	9.32	16.48	8.97	15.96	14.90
	TimesFM	19.56	0.58	24.01	35.02	10.44	6.13	98.39	72.89	6.03	21.57	86.78	11.84	14.76	16.95	13.02	19.43
	Time-MoE	16.63	0.52	22.62	19.76	9.34	10.87	74.78	48.78	2.10	10.93	20.90	7.82	15.68	4.98	11.12	16.20
	OmniAnomaly	25.35	0.64	27.17	74.51	14.32	6.20	91.29	45.55	2.40	16.37	29.26	31.57	18.58	28.07	37.44	42.97
	USAD	16.58	0.75	55.03	58.53	18.68	4.37	74.53	56.36	8.85	10.00	14.15	29.95	17.59	26.37	34.53	44.73
	TranAD	21.61	0.64	24.82	61.63	13.04	5.75	78.08	47.33	2.25	12.20	25.78	14.78	16.49	13.37	28.34	47.37
Aff.-F1	TFMAE	5.32	0.64	15.68	17.81	11.90	9.55	73.54	48.79	2.57	5.36	25.93	8.25	14.22	5.76	4.77	15.38
	DCdetector	5.83	0.59	16.60	14.03	12.32	9.37	74.16	46.66	1.53	3.23	10.17	7.01	14.49	4.21	4.66	15.04
	TS-Router	90.09	90.72	93.14	86.19	88.88	94.81	68.46	75.48	90.43	97.16	93.80	83.61	77.09	88.25	91.55	80.25
	TimeRCD	83.28	70.69	82.48	79.73	85.51	96.87	71.84	86.37	84.63	90.33	96.65	81.16	81.61	87.73	92.58	71.55
	DADA	89.37	67.66	86.56	95.40	69.79	65.18	98.77	76.89	72.21	93.92	92.20	76.57	81.27	76.92	83.74	76.18
	Chronos	90.12	67.89	86.66	93.63	69.72	67.89	96.85	91.96	74.35	90.98	96.34	75.52	70.88	72.22	75.31	70.43
	MOMENT	87.54	66.76	90.45	92.26	75.97	59.13	45.26	59.76	75.77	95.39	79.99	74.55	65.79	77.42	74.00	70.17
	TimesFM	81.88	66.95	79.73	90.49	69.88	67.14	97.53	89.08	70.03	78.97	91.28	20.35	71.24	45.44	62.85	44.37
	Time-MoE	76.34	67.23	80.51	80.50	71.19	60.98	63.28	54.68	73.56	80.25	69.70	69.85	54.74	74.38	69.97	64.37
	OmniAnomaly	80.32	67.35	92.35	86.30	78.16	61.26	75.24	50.73	73.53	78.02	71.31	83.15	58.17	91.38	85.82	73.39
F1_T	USAD	71.08	67.81	91.54	71.13	76.48	55.60	35.92	47.90	76.00	65.10	53.05	81.86	57.86	87.25	85.09	75.06
	TranAD	83.19	67.28	90.28	85.02	71.56	61.03	57.94	52.76	73.31	84.34	76.08	79.91	73.83	87.39	92.20	75.37
	TFMAE	78.25	67.50	75.99	76.91	70.30	68.17	56.39	62.83	70.60	80.25	76.87	75.70	70.07	75.36	70.85	75.72
	DCdetector	71.83	67.91	72.21	62.31	69.75	72.20	55.79	57.81	70.18	72.79	67.77	67.74	67.32	67.10	69.55	71.07
	TS-Router	48.63	36.62	56.81	84.18	30.26	58.43	17.26	34.38	47.83	57.87	73.58	43.30	27.90	43.61	52.26	45.04
	TimeRCD	28.44	1.81	38.85	35.87	28.47	69.43	31.73	65.89	34.30	35.04	85.86	42.47	37.98	33.74	53.91	30.28
	DADA	42.50	0.91	37.24	47.98	19.80	9.56	95.49	35.18	7.22	48.46	79.52	34.58	31.84	30.42	40.80	35.13
	Chronos	45.45	1.10	36.10	33.16	19.90	13.18	89.30	53.90	10.88	39.82	79.00	15.59	25.42	11.72	17.32	28.88
	MOMENT	33.15	0.80	52.27	63.66	19.91	9.54	18.04	17.47	13.02	41.98	11.69	25.97	27.77	17.93	28.68	28.76
	TimesFM	48.95	0.93	36.74	36.63	19.80	9.58	88.94	51.13	10.78	41.38	83.46	7.83	25.42	11.64	18.65	21.39
Std.-F1	Time-MoE	25.95	0.63	38.70	15.78	19.85	17.73	34.13	20.91	8.29	22.60	37.11	23.92	26.82	14.22	19.90	30.11
	OmniAnomaly	51.17	1.61	40.09	82.20	23.48	9.68	36.22	14.33	8.47	34.79	24.16	49.36	30.42	46.63	51.84	46.64
	USAD	20.99	4.07	61.46	70.64	28.23	9.54	16.86	20.85	14.63	14.18	9.35	48.71	28.96	43.94	50.41	50.41
	TranAD	22.63	1.65	37.28	69.97	22.36	9.57	16.73	13.51	7.75	20.94	8.41	39.42	25.49	29.12	37.98	49.58
	TFMAE	19.41	1.07	33.04	31.11	20.18	11.82	22.15	16.48	5.90	19.11	23.85	25.28	25.36	19.39	10.13	28.46
	DCdetector	6.61	1.32	32.72	29.21	21.13	10.53	16.07	16.40	6.62	7.32	6.81	23.24	25.34	15.73	9.47	28.64
	TS-Router	42.44	36.56	52.01	75.05	29.98	58.71	18.83	40.41	43.88	56.41	70.78	28.79	26.97	40.30	53.20	51.08
	TimeRCD	24.22	1.62	27.70	33.05	28.59	69.88	32.61	67.02	28.13	31.96	87.02	30.66	26.00	30.48	44.89	28.73
	DADA	32.76	0.80	26.91	48.24	15.99	2.69	95.59	28.18	3.36	45.06	79.30	22.13	24.07	26.75	34.98	34.78
	Chronos	32.69	0.99	26.22	33.54	17.47	8.74	89.41	40.52	8.21	34.58	78.89	11.63	22.27	9.62	17.50	24.03
Std.-F1	MOMENT	30.69	0.67	44.75	63.85	16.39	3.36	19.38	14.64	9.00	41.42	10.54	14.43	23.83	12.92	29.78	21.30
	TimesFM	34.28	0.83	26.46	38.15	16.73	2.96	89.13	40.08	7.86	38.50	84.44	5.75	22.18	10.46	18.65	22.84
	Time-MoE	26.52	0.45	26.20	11.47	12.16	17.73	34.32	16.38	4.09	20.09	27.50	12.85	24.80	9.01	21.62	23.58
	OmniAnomaly	47.05	1.44	28.81	74.03	23.50	0.43	38.59	12.65	5.11	29.57	21.40	39.10	30.43	40.50	57.06	55.93
	USAD	30.66	3.89	56.15	62.91	28.24	3.41	17.99	23.87	10.74	13.20	7.21	38.71	28.41	38.66	53.06	62.82
	TranAD	34.85	1.46	27.33	60.36	22.36	2.63	16.23	11.94	4.40	20.23	5.70	29.60	25.63	25.11	43.99	61.86
	TFMAE	9.48	0.97	23.78	19.74	20.14	11.87	23.32	14.88	2.83	15.53	20.50	15.68	25.39	12.58	9.16	27.08
	DCdetector	5.19	1.21	24.02	17.37	21.10	10.54	16.97	17.85	3.18	4.64	4.14	14.08	25.33	10.67	8.99	27.02

are also broadly distributed across datasets rather than arising from a single benchmark. Handcrafted representations such as Catch22, TSFresh, and BasicStats can be strong on particular datasets but exhibit larger variation across temporal domains. Chronos and MOMENT remain weaker on VUS-PR and Standard-F1.

Together with the NDCG and Hit values reported in Table 2, these results suggest that the advantage of the pretrained temporal representation is not limited to downstream score fusion. It also yields a more informative description of the target series for predicting specialist competence. This is consistent with using the representation to decide which specialist criterion to trust, rather than as a universal anomaly score.

Table 8: Representation ablation on eleven univariate TSB-AD benchmarks. All variants use the same MLP, 11-specialist pool, and Top-3 normalized mean fusion; only the router representation changes. TimeRCD remains frozen. Best results are in **bold** and second-best results are underlined.

Metric	Representation	Univariate datasets											Avg.
		IOPS	MGAB	NAB	NEK	Power	SED	Stock	TODS	UCR	WSD	YAHOO	
VUS-PR	TimeRCD	<u>41.49</u>	43.09	51.97	<u>76.92</u>	22.40	<u>65.42</u>	73.18	71.65	39.60	56.91	<u>77.32</u>	56.36
	Chronos	38.63	33.44	49.34	50.10	15.99	94.18	71.64	78.81	33.17	53.83	66.87	53.27
	MOMENT	40.73	33.44	46.83	33.93	20.56	94.18	87.24	76.35	<u>38.09</u>	<u>57.55</u>	63.87	<u>53.89</u>
	Catch22	44.58	9.46	42.75	38.31	<u>22.27</u>	94.18	76.57	84.05	34.11	48.86	54.39	49.96
	TSFresh	36.98	35.44	35.53	50.10	17.54	37.17	<u>88.43</u>	73.84	33.18	51.50	79.40	49.01
	BasicStats	40.65	<u>37.51</u>	43.69	42.82	20.93	6.52	90.20	72.17	27.98	41.45	76.36	45.48
	MiniRocket	32.08	12.56	<u>50.28</u>	77.04	14.72	61.63	78.24	<u>83.86</u>	32.26	57.57	64.19	51.31
Aff.-F1	TimeRCD	90.09	90.72	93.14	86.19	88.88	<u>94.81</u>	68.46	75.48	<u>90.43</u>	97.16	<u>93.80</u>	88.11
	Chronos	88.80	85.47	<u>93.38</u>	81.72	85.66	98.67	68.45	80.91	87.75	96.21	92.73	87.25
	MOMENT	91.69	85.47	93.07	76.63	<u>87.16</u>	98.67	81.16	80.00	90.56	<u>97.26</u>	91.79	88.50
	Catch22	89.75	74.09	91.02	80.67	78.37	98.67	68.70	78.61	88.65	93.45	89.16	84.65
	TSFresh	<u>92.18</u>	<u>88.27</u>	93.05	<u>84.74</u>	82.58	84.07	84.72	<u>80.58</u>	88.83	96.54	94.17	<u>88.16</u>
	BasicStats	93.03	87.51	92.57	70.62	83.15	68.21	83.69	75.94	83.76	94.91	94.17	84.32
	MiniRocket	88.08	77.33	93.93	81.91	86.08	91.37	69.65	79.55	88.60	97.33	91.19	85.91
F1 _T	TimeRCD	48.63	36.62	56.81	84.18	30.26	58.43	17.26	34.38	47.83	57.87	73.58	49.62
	Chronos	44.97	36.67	53.78	60.33	22.77	75.27	17.46	43.79	39.91	57.75	52.47	45.93
	MOMENT	<u>49.51</u>	36.67	52.20	49.99	28.94	75.27	56.67	42.05	<u>46.09</u>	60.13	49.59	49.74
	Catch22	49.37	26.12	50.55	49.94	27.99	75.27	18.73	38.57	40.73	51.01	36.82	42.28
	TSFresh	49.21	32.95	41.77	56.30	24.50	29.90	<u>66.62</u>	39.60	42.03	56.68	<u>78.91</u>	47.13
	BasicStats	52.98	32.47	50.97	54.98	25.48	9.79	67.71	30.19	35.89	50.33	79.76	44.60
	MiniRocket	45.30	33.56	<u>56.06</u>	<u>80.46</u>	22.37	48.85	20.12	38.74	39.99	<u>58.78</u>	46.73	44.63
Std.-F1	TimeRCD	<u>42.44</u>	<u>36.56</u>	52.01	<u>75.05</u>	29.98	<u>58.71</u>	18.83	40.41	43.88	56.41	70.78	47.73
	Chronos	38.11	36.68	49.33	49.39	22.63	75.69	19.12	37.97	35.49	56.28	49.95	42.79
	MOMENT	40.24	36.68	46.04	36.81	<u>28.61</u>	75.69	57.75	41.62	<u>41.74</u>	57.93	46.98	<u>46.37</u>
	Catch22	44.02	26.20	45.34	36.98	28.03	75.69	20.50	44.76	35.78	50.24	33.91	40.13
	TSFresh	38.96	33.05	36.14	44.56	24.53	29.98	<u>67.34</u>	29.32	36.97	54.25	<u>76.34</u>	42.86
	BasicStats	41.30	32.46	43.93	46.40	25.84	9.80	68.39	30.98	29.89	48.22	77.21	41.31
	MiniRocket	41.96	33.62	<u>50.05</u>	75.49	22.15	49.14	22.09	46.45	36.78	<u>56.94</u>	44.19	43.53

B.2.3 SPECIALIST COMPLEMENTARITY

A key premise of TS-Router is that heterogeneous anomaly detectors provide complementary inductive biases rather than forming a uniformly ordered set of strong and weak models. Table 9 reports the complete performance of all eleven specialists on the univariate benchmarks. The relative strengths of individual specialists vary substantially across datasets. For example, subspace-based methods are particularly effective on NEK, neighborhood-based specialists perform strongly on MGAB and SED, KMeansAD is competitive on Power, and SR is especially strong on Stock and YAHOO. No individual specialist consistently dominates across targets.

This complementarity also appears at the aggregate level. The best fixed individual specialist achieves average scores of 39.68, 84.34, 41.01, and 36.43 under VUS-PR, Affiliation-F1, F1_T, and Standard-F1, respectively, whereas TS-Router reaches 56.36, 88.11, 49.62, and 47.73. Thus, the advantage of TS-Router cannot be explained by repeatedly selecting a single globally strong detector. Instead, routing exploits changes in relative specialist competence across temporal contexts.

We additionally report a hindsight Oracle that selects the best individual specialist separately for each target using its ground-truth detection performance. Together with Best Fixed, it illustrates the scope for individual-specialist adaptation rather than an upper bound on fused detection.

B.2.4 EFFECT OF TOP- k SELECTION

We sweep the number of selected specialists k from 1 to 10 while keeping the 11-specialist pool and the VUS-PR routing head fixed. Figure 3 reports dataset-averaged scores over the eleven univariate benchmarks. VUS-PR increases from 49.00 at $k = 1$; after a small selection budget, the scores decline as k grows toward 10. This sweep evaluates the empirical budget tradeoff in the com-

Table 9: Specialist complementarity on eleven univariate TSB-AD benchmarks. TS-Router uses Top-3 normalized mean fusion, while Oracle denotes hindsight per-target selection of the best individual specialist. Oracle is an individual-specialist reference rather than an upper bound on the fused TS-Router prediction. Best and second-best non-oracle results are in **bold** and underlined, respectively.

Metric	Method	Univariate datasets											Avg.
		IOPS	MGAB	NAB	NEK	Power	SED	Stock	TODS	UCR	WSD	YAHOO	
VUS-PR	TS-Router	41.49	43.09	51.97	76.92	22.40	65.42	73.18	<u>71.65</u>	39.60	56.91	77.32	56.36
	PCA	23.21	0.60	45.77	<u>78.45</u>	10.49	3.77	<u>80.92</u>	54.01	13.40	16.95	21.15	31.70
	Sub-PCA	20.35	0.66	<u>49.82</u>	83.68	10.49	4.00	77.81	52.82	13.53	14.09	19.41	31.51
	Sub-HBOS	4.97	0.50	33.26	19.43	15.69	63.09	64.60	64.54	15.07	2.21	12.08	26.86
	POLY	27.51	0.70	46.91	55.64	9.30	7.88	79.01	57.39	15.80	35.24	27.89	33.02
	Sub-KNN	8.93	24.70	34.07	31.82	<u>24.97</u>	88.02	70.05	67.33	34.03	10.27	28.83	38.46
	Sub-LOF	<u>27.62</u>	48.14	39.92	41.83	<u>17.46</u>	24.70	74.86	53.07	<u>35.25</u>	<u>46.18</u>	27.45	<u>39.68</u>
	KMeansAD	6.71	0.97	34.96	27.06	37.05	<u>84.19</u>	71.34	67.12	34.77	9.57	46.58	38.21
	OCSVM	7.81	0.61	27.17	18.30	20.20	9.81	68.43	76.08	9.79	3.74	41.75	25.79
	Sub-OCSVM	7.11	0.93	30.43	25.79	19.83	6.97	70.42	68.73	9.87	3.46	24.03	24.32
	Sub-IForest	14.38	0.56	45.13	68.04	9.79	55.74	75.94	48.35	8.12	6.39	15.64	31.64
	SR	24.72	0.79	24.69	48.36	12.39	7.80	99.65	61.66	7.59	23.07	<u>71.51</u>	34.75
	Oracle	45.14	48.14	71.69	83.73	37.05	88.85	99.65	80.44	54.85	60.11	78.94	68.05
Aff.-F1	TS-Router	<u>90.09</u>	<u>90.72</u>	93.14	86.19	88.88	94.81	68.46	<u>75.48</u>	90.43	97.16	93.80	88.11
	PCA	78.75	67.32	92.46	<u>86.48</u>	72.59	67.14	70.65	71.97	80.41	78.76	74.97	76.50
	Sub-PCA	78.78	67.42	<u>93.09</u>	87.36	72.59	67.14	<u>71.30</u>	71.61	80.32	77.86	73.99	76.50
	Sub-HBOS	68.78	67.17	84.88	70.59	<u>78.27</u>	92.35	68.01	72.67	81.25	70.65	72.28	75.17
	POLY	80.95	68.86	88.90	82.73	71.17	64.89	69.18	69.43	82.68	83.74	79.27	76.53
	Sub-KNN	70.48	79.28	82.95	68.42	86.52	98.64	68.24	73.99	87.52	74.67	85.89	79.69
	Sub-LOF	85.13	94.92	91.33	79.52	87.97	82.36	67.92	71.63	<u>90.29</u>	<u>95.71</u>	80.91	84.34
	KMeansAD	70.23	69.27	82.46	67.44	<u>88.58</u>	<u>97.32</u>	67.97	72.86	85.08	73.74	87.13	78.37
	OCSVM	71.21	67.11	82.49	67.16	76.81	72.12	68.40	75.45	79.06	72.81	88.19	74.62
	Sub-OCSVM	70.75	68.41	84.94	71.69	85.47	70.54	68.62	72.88	78.60	72.91	81.64	75.13
	Sub-IForest	73.75	67.11	91.02	79.50	85.92	91.05	69.31	68.76	77.49	75.01	72.01	77.36
	SR	92.04	69.68	86.80	85.34	75.92	68.22	99.33	83.39	74.23	92.14	<u>91.61</u>	83.52
	Oracle	94.44	94.92	96.78	87.97	88.58	98.64	99.33	85.01	97.14	98.75	95.58	94.29
F1 _T	TS-Router	48.63	<u>36.62</u>	<u>56.81</u>	84.18	<u>30.26</u>	58.43	17.26	<u>34.38</u>	47.83	57.87	<u>73.58</u>	49.62
	PCA	31.83	0.99	54.09	<u>85.76</u>	20.30	9.54	20.13	18.36	19.38	24.41	12.93	27.07
	Sub-PCA	27.70	2.51	57.88	88.26	20.30	9.55	<u>20.56</u>	17.91	19.66	21.69	12.34	27.12
	Sub-HBOS	6.70	0.76	44.63	33.91	24.23	49.93	16.30	21.94	21.89	5.48	8.71	21.32
	POLY	32.07	1.30	51.62	68.16	16.95	14.82	19.71	23.36	20.62	35.42	10.75	26.80
	Sub-KNN	7.06	34.69	46.55	45.63	28.20	<u>63.62</u>	16.58	26.45	40.91	11.72	13.50	30.45
	Sub-LOF	33.36	39.61	46.75	54.87	23.59	17.86	18.05	16.42	<u>45.18</u>	<u>47.80</u>	13.06	32.41
	KMeansAD	7.73	8.29	47.88	49.35	36.65	70.88	16.55	28.73	41.14	12.52	36.25	32.36
	OCSVM	8.57	1.22	39.51	35.56	27.48	17.92	16.73	28.60	15.07	7.69	16.90	19.57
	Sub-OCSVM	6.52	7.33	41.83	38.92	26.83	16.14	17.45	26.29	14.85	7.11	11.65	19.54
	Sub-IForest	15.79	0.84	50.73	75.16	19.79	44.83	18.19	14.28	16.26	15.51	12.23	25.78
	SR	<u>47.06</u>	1.42	38.12	54.60	20.67	9.59	97.94	45.90	13.79	40.84	81.17	<u>41.01</u>
	Oracle	57.01	39.81	75.53	88.37	36.65	70.88	97.94	56.57	63.52	62.87	83.29	66.59
Std.-F1	TS-Router	42.44	<u>36.56</u>	<u>52.01</u>	75.05	<u>29.98</u>	58.71	18.83	40.41	43.88	56.41	<u>70.78</u>	47.73
	PCA	32.98	0.86	49.33	<u>75.76</u>	20.30	9.43	22.06	19.74	15.61	25.18	10.50	25.61
	Sub-PCA	29.43	2.39	53.48	80.30	20.30	9.50	<u>22.52</u>	19.38	15.79	21.48	9.93	25.86
	Sub-HBOS	5.73	0.67	37.70	23.45	24.16	50.08	17.72	30.86	17.05	3.80	5.43	19.70
	POLY	28.95	1.14	50.18	58.50	16.95	14.82	21.65	29.39	17.87	36.43	8.87	25.89
	Sub-KNN	7.26	34.73	40.68	35.78	<u>28.22</u>	<u>63.81</u>	18.11	<u>43.74</u>	36.49	7.98	10.92	29.79
	Sub-LOF	27.55	39.60	40.91	43.13	23.46	17.87	19.81	18.98	<u>40.48</u>	<u>45.26</u>	10.93	29.82
	KMeansAD	8.99	8.22	42.37	36.16	36.65	70.93	17.93	44.40	36.97	9.63	34.10	31.49
	OCSVM	8.16	1.11	31.91	23.81	27.51	17.92	17.96	41.41	11.67	4.80	14.22	18.23
	Sub-OCSVM	5.78	7.24	34.61	26.83	26.85	16.15	18.61	41.28	11.85	4.37	8.88	18.40
	Sub-IForest	25.02	0.73	45.35	64.21	19.82	45.25	19.66	14.43	11.56	11.75	9.05	24.26
	SR	<u>33.82</u>	1.28	29.96	47.42	21.19	9.57	97.92	33.64	10.28	37.29	78.41	<u>36.43</u>
	Oracle	51.66	39.82	72.51	80.40	36.65	70.93	97.92	58.17	59.26	62.24	80.97	64.59

plete selection-and-fusion pipeline. Theorem 1 compares set competence at a fixed k , rather than predicting monotonic improvement of fused detection scores as the budget increases.

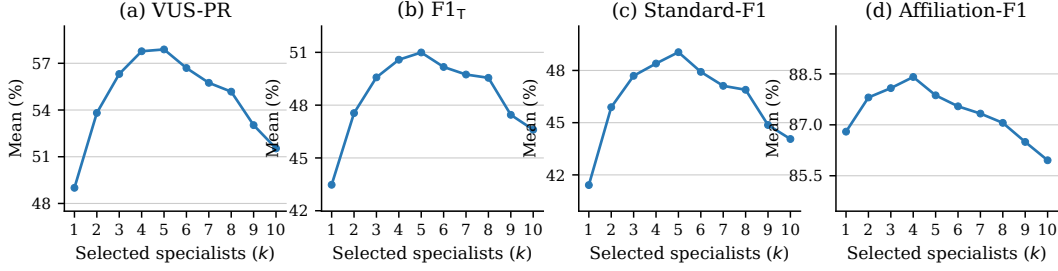


Figure 3: Effect of the number of selected specialists k with the 11-specialist pool. Each panel reports the unweighted mean over eleven univariate benchmarks; the panels show VUS-PR, $F1_T$, Standard-F1, and Affiliation-F1.

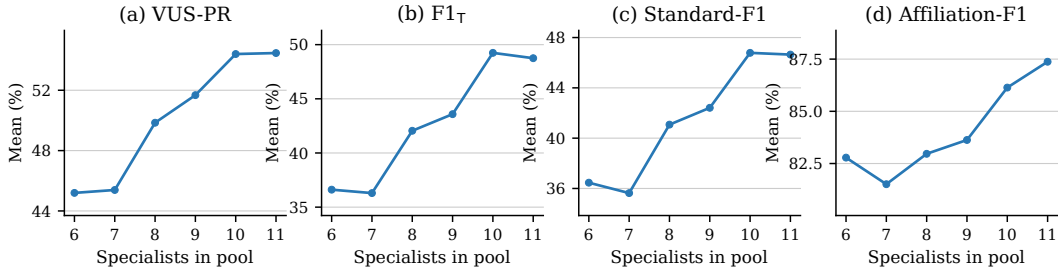


Figure 4: Effect of specialist-pool size with Top-3 selection. Each panel reports the unweighted mean over eleven univariate benchmarks; the panels show VUS-PR, $F1_T$, Standard-F1, and Affiliation-F1. The router is retrained for each pool size.

B.2.5 EFFECT OF SPECIALIST-POOL SIZE

We next vary the number of available specialists from 6 to 11. For each pool size, we retrain the router on the corresponding specialist pool and evaluate with Top-3 selection and the VUS-PR routing head. Figure 4 shows that enlarging the pool improves detection.

B.2.6 ROUTER-HEAD ABLATION

We evaluate four routers that differ only in the metric used to construct the competence target: VUS-PR, Affiliation-F1, $F1_T$ and Standard-F1. The specialist pool, frozen TimeRCD representation, Top-3 selection, and test set remain fixed. Each setting is evaluated on the same eleven univariate benchmarks as Table 2. Table 10 reports the downstream detection metrics, including the metric used as the routing head.

The $F1_T$ head gives the highest mean score under all five reported detection metrics, including VUS-PR (58.24) and Affiliation-F1 (89.17). The VUS-PR head remains close on VUS-PR (56.36), while the Standard-F1 head has the lowest VUS-PR, VUS-ROC and Affiliation-F1 values in this comparison. These results show that the supervision head changes the selected specialist rankings and can affect metrics beyond the head’s own target.

B.2.7 ADDITIONAL ALGORITHM-SELECTION TRANSFER SETTINGS

We further compare TS-Router with classical algorithm-selection approaches under three complementary transfer settings. These experiments examine whether specialist competence can be transferred when selectors are provided with different forms of historical performance information. The three settings are not intended to impose identical supervision budgets. Instead, they progressively test selection under synthetic competence histories, limited labeled support from the target distribution, and labeled histories from other real datasets. In all settings, TS-Router keeps the same routing

Table 10: Top-3 routing with different competence-supervision heads. Values are dataset-averaged percentages over eleven univariate benchmarks. The pool size is 11 and the router is trained with the indicated head.

Routing head	VUS-PR	VUS-ROC	F1 _T	Std.-F1	Aff.-F1
VUS-PR	56.36	88.30	49.62	47.73	88.11
Affiliation-F1	56.27	89.24	49.17	46.83	88.49
F1 _T	58.24	89.29	53.03	49.96	89.17
Standard-F1	55.28	87.81	50.00	47.77	87.83

Table 11: Algorithm-selection transfer on eleven univariate TSB-AD benchmarks. Classical selectors are trained on Syn-RCD using the common 11-specialist pool. TS-Router uses its fixed routing policy learned from the same synthetic competence supervision. All selectors use Top-3 fusion of the predicted specialists. Best results are in **bold** and second-best results are underlined.

Metric	Selector	Univariate datasets											Avg.
		IOPS	MGAB	NAB	NEK	Power	SED	Stock	TODS	UCR	WSD	YAHOO	
VUS-PR	TS-Router	41.49	43.09	51.97	76.92	22.40	65.42	73.18	71.65	39.60	56.91	77.32	56.36
	SATzilla	41.77	<u>33.44</u>	41.71	53.12	22.40	74.65	70.43	72.73	<u>41.82</u>	58.30	<u>59.20</u>	<u>51.78</u>
	ARGOSMART	44.02	9.46	<u>44.73</u>	31.94	22.36	11.94	82.21	70.71	34.24	54.88	53.37	41.81
	MetaOD	42.21	<u>33.44</u>	40.23	<u>57.82</u>	22.40	33.96	70.20	72.73	<u>41.82</u>	56.72	57.89	48.13
	ISAC	43.02	<u>33.44</u>	41.84	39.78	22.40	74.65	76.03	72.72	40.42	58.08	57.09	50.86
	MSAD	36.70	<u>33.44</u>	42.09	42.31	22.40	<u>65.87</u>	70.74	72.42	40.98	52.49	56.72	48.74
	UReg	42.12	<u>33.44</u>	41.57	40.33	22.40	74.65	70.48	72.73	41.83	51.18	53.89	49.51
Aff.-F1	TS-Router	90.09	90.72	93.14	86.19	88.88	94.81	68.46	75.48	90.43	97.16	93.80	88.11
	SATzilla	90.42	<u>85.47</u>	91.38	81.61	88.88	96.78	67.99	77.48	<u>92.06</u>	97.58	<u>90.96</u>	<u>87.33</u>
	ARGOSMART	<u>92.45</u>	74.09	<u>92.75</u>	75.75	84.13	71.40	73.97	75.77	89.40	96.41	87.86	83.09
	MetaOD	90.46	<u>85.47</u>	91.59	<u>81.97</u>	88.88	85.87	67.99	77.48	<u>92.06</u>	97.11	90.70	86.32
	ISAC	93.20	<u>85.47</u>	92.15	76.88	88.88	96.78	68.25	77.19	91.80	97.44	89.19	87.02
	MSAD	88.48	<u>85.47</u>	92.31	77.11	88.88	<u>95.03</u>	67.99	77.13	91.82	95.77	90.54	86.41
	UReg	86.39	<u>85.47</u>	91.76	77.10	88.88	96.78	67.99	77.48	92.11	94.75	89.71	86.22
F1 _T	TS-Router	48.63	<u>36.62</u>	56.81	84.18	30.26	58.43	17.26	34.38	47.83	57.87	73.58	49.62
	SATzilla	49.56	36.67	48.45	62.45	30.26	68.11	16.55	<u>34.32</u>	<u>49.33</u>	60.88	44.54	<u>45.56</u>
	ARGOSMART	49.83	26.12	<u>50.63</u>	47.34	<u>27.12</u>	15.80	35.48	30.10	42.14	58.05	44.55	38.83
	MetaOD	<u>50.20</u>	36.67	47.34	<u>66.39</u>	30.26	28.79	16.46	<u>34.32</u>	<u>49.33</u>	59.79	41.95	41.96
	ISAC	50.63	36.67	47.21	54.62	30.26	68.11	18.66	34.27	48.36	59.83	48.66	45.21
	MSAD	42.76	36.67	46.34	56.05	30.26	<u>58.95</u>	16.49	33.53	48.98	54.83	37.84	42.06
	UReg	43.45	36.67	48.29	53.95	30.26	68.11	16.46	<u>34.32</u>	49.39	53.22	35.23	42.67
Std.-F1	TS-Router	42.44	<u>36.56</u>	52.01	75.05	29.98	58.71	18.83	40.41	43.88	56.41	70.78	47.73
	SATzilla	41.94	36.68	43.11	52.50	29.98	68.39	18.13	<u>36.75</u>	<u>45.45</u>	58.58	41.13	<u>42.97</u>
	ARGOSMART	41.00	26.20	<u>44.37</u>	34.55	<u>27.52</u>	15.77	36.92	36.00	38.30	55.77	39.97	36.03
	MetaOD	<u>42.47</u>	36.68	41.45	<u>56.43</u>	29.98	28.88	18.03	<u>36.75</u>	<u>45.45</u>	57.50	38.57	39.29
	ISAC	41.29	36.68	41.41	41.46	29.98	68.39	<u>20.48</u>	36.71	44.25	57.63	<u>44.10</u>	42.04
	MSAD	37.75	36.68	40.94	42.89	29.98	<u>59.18</u>	18.06	35.56	44.39	52.42	34.82	39.33
	UReg	43.35	36.68	42.81	40.80	29.98	68.39	18.03	<u>36.75</u>	45.49	51.58	32.19	40.55

policy learned from Syn-RCD and receives no additional adaptation from the real support or target sets.

Synthetic-to-Real Selection. We first consider the setting most directly aligned with the training protocol of TS-Router. Classical algorithm selectors are trained on Syn-RCD using specialist competence computed from the same eleven-specialist pool, and are then transferred directly to the eleven real univariate benchmarks. Neither approach receives real target anomaly labels for routing adaptation.

Table 11 reports this comparison under a shared Top-3 fusion protocol. TS-Router still achieves the highest average performance under all four detection metrics. Classical selectors remain competitive on particular datasets, indicating that synthetic competence histories contain useful transferable information, but their relative performance varies substantially across targets.

In-Distribution Selection with Limited Support. We next consider a substantially more favorable setting for classical algorithm selection, restricted to the official TSB-AD Tuning split. This split

Table 12: In-distribution algorithm selection on the official TSB-AD Tuning subset. Evaluation uses the thirty genuinely univariate Tuning series from ten collections; Power does not appear in the Tuning split. Classical selectors are adapted from a small labeled support set across ten random trials, whereas TS-Router uses a fixed routing policy without support-set adaptation. Best results are in **bold** and second-best results are underlined.

Metric	Method	Univariate datasets										Avg.
		IOPS	MGAB	NAB	NEK	SED	Stock	TODS	UCR	WSD	YAHOO	
VUS-PR	TS-Router	28.72	<u>45.61</u>	57.45	94.24	63.78	52.84	87.54	29.67	45.05	69.09	57.40
	SATzilla	20.45	48.52	<u>43.74</u>	<u>76.40</u>	<u>77.57</u>	90.36	73.87	<u>38.32</u>	35.01	45.81	55.00
	ARGOSMART	<u>23.07</u>	48.52	42.86	41.35	<u>77.57</u>	<u>94.57</u>	<u>78.03</u>	<u>25.53</u>	<u>38.48</u>	49.53	51.95
	MetaOD	19.58	48.52	39.00	<u>76.40</u>	<u>77.57</u>	90.36	77.94	39.62	35.01	<u>55.13</u>	<u>55.91</u>
	ISAC	18.96	42.10	42.93	<u>36.70</u>	<u>36.09</u>	52.00	64.30	31.10	33.26	<u>19.34</u>	<u>37.68</u>
	MSAD	20.06	48.52	34.97	70.09	<u>77.57</u>	99.21	65.82	33.31	36.73	47.28	53.36
	UReg	19.38	48.52	39.95	34.09	89.25	52.09	62.08	32.39	35.77	16.16	42.97
Aff.-F1	TS-Router	<u>86.08</u>	90.12	94.44	99.16	93.25	68.30	<u>82.30</u>	83.45	98.96	92.82	88.89
	SATzilla	85.15	94.46	<u>89.02</u>	91.91	<u>96.18</u>	94.29	81.59	<u>88.08</u>	91.18	89.00	<u>90.09</u>
	ARGOSMART	87.88	94.46	83.80	84.29	<u>96.18</u>	<u>96.12</u>	81.99	83.44	<u>96.21</u>	90.91	89.53
	MetaOD	83.23	94.46	87.40	91.91	<u>96.18</u>	<u>94.29</u>	82.43	88.44	<u>91.18</u>	<u>91.05</u>	90.06
	ISAC	82.40	<u>90.92</u>	86.43	78.10	82.92	67.72	79.72	84.75	90.78	86.13	82.99
	MSAD	86.06	94.46	87.60	<u>94.27</u>	<u>96.18</u>	99.46	80.11	87.22	95.84	88.00	90.92
	UReg	80.82	94.46	88.55	<u>77.81</u>	98.60	67.52	79.69	86.16	95.14	83.36	85.21
F1 _T	TS-Router	44.30	36.99	62.44	87.59	47.05	15.87	<u>40.34</u>	26.97	62.03	64.34	48.79
	SATzilla	25.47	43.95	50.67	<u>73.75</u>	<u>55.04</u>	85.13	37.60	<u>38.34</u>	41.80	37.58	<u>48.93</u>
	ARGOSMART	29.88	43.95	48.16	<u>43.53</u>	<u>55.04</u>	<u>89.69</u>	40.50	29.43	<u>53.87</u>	34.76	46.88
	MetaOD	17.37	43.95	46.84	<u>73.75</u>	<u>55.04</u>	85.13	40.26	38.84	41.80	<u>44.83</u>	48.78
	ISAC	23.95	<u>42.14</u>	<u>52.81</u>	37.89	32.67	17.79	30.14	31.51	44.69	7.44	32.10
	MSAD	<u>30.24</u>	43.95	44.51	65.55	<u>55.04</u>	98.66	32.26	35.71	51.47	40.26	49.76
	UReg	17.26	43.95	47.95	34.77	65.87	17.60	27.63	34.00	52.48	9.14	35.06
Std.-F1	TS-Router	43.85	37.09	57.00	86.73	47.35	17.33	<u>40.32</u>	24.66	62.06	63.74	48.01
	SATzilla	25.65	43.95	45.23	<u>74.20</u>	<u>55.31</u>	85.34	<u>37.48</u>	<u>34.39</u>	42.69	37.95	<u>48.22</u>
	ARGOSMART	30.18	43.95	46.15	44.48	<u>55.31</u>	<u>89.80</u>	40.71	25.48	<u>54.68</u>	34.75	46.55
	MetaOD	17.24	43.95	41.20	<u>74.20</u>	<u>55.31</u>	85.34	40.28	35.25	42.69	<u>45.39</u>	48.08
	ISAC	24.12	<u>42.19</u>	<u>46.29</u>	41.13	32.79	19.52	30.01	26.71	45.81	7.36	31.59
	MSAD	<u>30.56</u>	43.95	38.00	65.42	<u>55.31</u>	98.58	31.91	31.94	52.44	40.46	48.86
	UReg	17.32	43.95	41.10	38.12	66.12	19.40	27.47	28.63	53.69	9.02	34.48

provides the in-distribution subset used to develop detectors and selectors; after excluding decomposed multivariate series, it contains thirty genuinely univariate series from ten collections (IOPS, MGAB, NAB, NEK, SED, Stock, TODS, UCR, WSD, and YAHOO). Power does not appear in the Tuning list and is therefore omitted. Classical selectors may use a labeled support set of approximately 10% of series, drawn over ten random trials, to infer which specialists tend to perform well within that target distribution. We report their test-set scores on the Tuning series. TS-Router receives no support-set adaptation and is evaluated with exactly the same fixed policy as in the main experiments.

As shown in Table 12, access to in-distribution support substantially narrows the gap for classical selectors. MetaOD reaches an average VUS-PR of 55.91, compared with 57.40 for TS-Router. MSAD attains the strongest Affiliation-F1, F1_T, and Standard-F1 averages (90.92, 49.76, and 48.86), slightly above TS-Router at 88.89, 48.79, and 48.01. This result is informative because the classical selectors are now given labeled target-distribution history that TS-Router does not use. Even under this stronger supervision condition, the fixed TS-Router policy remains competitive across all four metrics, suggesting that the pretrained competence representation captures information that otherwise requires explicit target-specific selection history.

Leave-One-Dataset-Out Transfer. Finally, we examine cross-dataset transfer using real historical specialist performance. For each of the eleven univariate benchmarks, the complete dataset is held out as an unseen target, while classical selectors are trained on specialist-performance records from

Table 13: Leave-one-dataset-out transfer across eleven univariate benchmarks. Classical selectors are trained on the other ten real benchmark collections and deploy their top-ranked specialist (Top-1). TS-Router uses its fixed routing policy learned on Syn-RCD and default Top-3 normalized mean fusion, without real-data retraining of the router. All methods are evaluated on the entirely held-out target dataset. Best results are in **bold** and second-best results are underlined.

Metric	Method	Held-out target dataset											Avg.
		IOPS	MGAB	NAB	NEK	Power	SED	Stock	TODS	UCR	WSD	YAHOO	
VUS-PR	TS-Router	41.49	43.09	51.97	76.92	22.40	65.42	73.18	71.65	39.60	56.91	77.32	56.36
	SATzilla	29.32	0.97	38.66	39.31	<u>24.97</u>	7.88	<u>77.81</u>	62.13	20.21	43.48	29.34	34.01
	ARGOSMART	27.68	<u>24.70</u>	32.40	<u>52.31</u>	9.30	<u>24.70</u>	76.93	57.39	20.83	44.15	<u>36.26</u>	36.97
	MetaOD	27.79	0.97	40.12	39.31	<u>24.97</u>	7.88	<u>77.81</u>	63.46	20.43	44.46	29.37	34.23
	ISAC	<u>31.77</u>	<u>24.70</u>	41.41	41.83	37.05	7.88	78.19	<u>64.33</u>	<u>25.04</u>	40.10	30.47	<u>38.43</u>
	MSAD	28.16	0.97	40.03	40.37	9.30	<u>24.70</u>	74.98	63.76	14.51	46.07	29.46	33.85
	UReg	29.31	0.97	<u>42.83</u>	41.83	17.46	<u>24.70</u>	77.10	53.30	12.03	<u>46.77</u>	27.18	33.95
Aff.-F1	TS-Router	90.09	90.72	93.14	86.19	88.88	94.81	68.46	75.48	90.43	97.16	93.80	88.11
	SATzilla	86.96	69.27	87.37	77.77	86.52	64.89	71.30	74.20	80.48	93.63	81.50	79.44
	ARGOSMART	86.34	<u>79.28</u>	84.17	74.73	71.17	<u>82.36</u>	68.99	69.43	81.76	94.43	<u>82.65</u>	79.57
	MetaOD	84.96	<u>69.27</u>	88.36	77.77	86.52	64.89	71.30	73.08	81.00	94.04	81.59	79.34
	ISAC	<u>88.05</u>	<u>79.28</u>	91.35	<u>79.52</u>	<u>88.58</u>	64.89	68.93	73.40	<u>82.86</u>	92.70	81.36	<u>80.99</u>
	MSAD	86.49	69.27	90.61	78.49	71.17	<u>82.36</u>	67.73	<u>74.49</u>	<u>78.61</u>	<u>95.57</u>	81.48	79.66
	UReg	85.62	69.27	<u>92.12</u>	<u>79.52</u>	87.97	<u>82.36</u>	68.43	71.62	76.23	95.45	80.92	80.86
F1 _T	TS-Router	48.63	36.62	56.81	84.18	<u>30.26</u>	58.43	17.26	34.38	47.83	57.87	73.58	49.62
	SATzilla	33.93	8.29	<u>50.00</u>	58.02	28.20	14.82	20.56	<u>28.38</u>	25.49	41.98	15.24	29.54
	ARGOSMART	33.01	<u>34.69</u>	41.08	<u>63.95</u>	16.95	<u>17.86</u>	18.75	23.36	27.31	45.83	<u>23.26</u>	31.46
	MetaOD	34.68	8.29	49.05	58.02	28.20	14.82	20.56	25.04	25.96	43.62	14.96	29.38
	ISAC	<u>39.96</u>	<u>34.69</u>	47.97	54.87	36.65	14.82	<u>19.07</u>	25.75	<u>31.73</u>	43.98	15.33	<u>33.17</u>
	MSAD	34.15	8.29	47.71	53.72	16.95	<u>17.86</u>	18.01	27.43	20.69	47.96	14.71	27.95
	UReg	33.34	8.29	48.04	54.87	23.59	<u>17.86</u>	18.76	16.66	19.08	<u>48.10</u>	12.82	27.40
Std.-F1	TS-Router	42.44	36.56	52.01	75.05	<u>29.98</u>	58.71	18.83	40.41	43.88	56.41	70.78	47.73
	SATzilla	29.63	8.22	42.78	43.26	28.22	14.82	22.52	38.81	22.32	40.57	13.22	27.67
	ARGOSMART	28.33	<u>34.73</u>	35.74	<u>52.80</u>	16.95	<u>17.87</u>	20.56	29.39	23.91	43.48	<u>21.15</u>	29.54
	MetaOD	29.75	8.22	42.54	43.26	28.22	14.82	22.52	<u>39.00</u>	22.74	42.44	12.79	27.85
	ISAC	<u>33.46</u>	<u>34.73</u>	42.38	43.13	36.65	14.82	<u>20.88</u>	<u>36.48</u>	<u>27.80</u>	42.07	13.37	<u>31.43</u>
	MSAD	28.36	8.22	41.94	41.75	16.95	<u>17.87</u>	19.81	33.08	17.55	45.32	12.43	25.75
	UReg	28.52	8.22	<u>42.97</u>	43.13	23.46	<u>17.87</u>	20.65	19.58	15.54	<u>45.55</u>	10.73	25.11

the remaining ten datasets. Each classical selector deploys its top-ranked specialist (Top-1), while TS-Router uses its default Top-3 normalized mean fusion. TS-Router retains its routing policy learned on Syn-RCD, without real-data retraining or target-specific adaptation of the router. This comparison evaluates the complete detection pipelines under cross-dataset transfer.

Table 13 shows that TS-Router achieves the highest dataset-averaged performance across all four metrics. The strongest classical-selector averages are 38.43, 80.99, 33.17, and 31.43 under VUS-PR, Affiliation-F1, F1_T, and Standard-F1, respectively, compared with 56.36, 88.11, 49.62, and 47.73 for TS-Router.

Moreover, the Router Top-1 variant reported in Table 2 also exceeds the strongest classical-selector average under each of the four metrics, showing that the advantage persists with single-specialist selection. These findings support the cross-dataset transferability of a competence-routing policy learned from pretrained temporal representations and synthetic supervision, without retraining the router on real-world detector-performance histories.